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Research Paper

Digital twin-enabled simulation system for NATM tunnel construction process considering multi-source uncertainty

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Abstract

During the construction of new Austrian tunnelling method (NATM) tunnels, inherent multi-source uncertainties may lead to substantial economic losses and project delays. To address this issue, a simulation system enabled by digital twin (DT) technology was developed. Guided by the concept of five-dimensional DT, a four-layer system architecture was established, including perception, transmission, modelling, and alternation layers. An integrated discrete event simulation (DES) model was then developed as the core of the system using structured system analysis. The model accounts for both geological and construction-related uncertainties, including routine delays and accident recovery durations. Geological uncertainties are represented through a stochastic geological modelling approach based on the Markov random field (MRF). Construction-related uncertainties are addressed using a tunnel vulnerability evaluation method based on the analytic hierarchy process – decision-making trial and evaluation laboratory (AHP–DEMATEL) method, along with the fitting of delay probability density functions. Finally, the proposed system was validated and evaluated in a real-world tunnel project. The results demonstrate that the system effectively analyzes multiple uncertainties, accurately estimates the construction schedule, enhances communication with the construction site, and ensures the safe and efficient completion of the project.

Keywords: NATM tunnelling; Digital twin; Process simulation; Markov random field; Tunnelling uncertainty

1 Introduction

1.1 Background

The new Austrian tunnelling method (NATM), due to its extensive applicability and economic advantages, has become a principal tunnelling methodology (Rabcewicz, 1964). However, tunnel construction is inherently influenced by multifaceted uncertainties, which pose significant challenges to project execution and safety.

Liu et al. (2018) reported that 60% of accidents result from human error, 10% from natural causes, and 30% from their combination. Sousa and Einstein (2021) categorized tunnel construction uncertainties into internal and external types. Internal uncertainties primarily involve conventional delay factors, such as variability in site organization and management, worker skill levels, and health conditions (Ye et al., 2023). For instance, the Heathrow Express Rail Link project in the UK faced a three-month delay due to human error. Although frequent, these factors typically have a limited impact and follow continuous probability distributions (Yu et al., 2017). In contrast, severe uncertainties relate to construction accidents such as rockfalls, collapses, floods, spalling, and excessive deformation (Sousa & Einstein, 2021). Despite their lower frequency, these incidents pose substantial safety and economic risks, exemplified by the 14 000 m^3 collapse in Nepal's Khimti I Hydropower Project.

Therefore, developing an effective approach to analyze these uncertainties is essential for advancing construction

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Nomenclature

$\boldsymbol{A}$	Direct influence matrix with internal values b_{ij}
B_e	Boolean variable
B_i	Being influenced degree of evaluation factors
C	Set of potential groups c_i
c_i	Potential groups formed by e_i and e_i^n
D_i	Influence degree of evaluation factors
D_j	Joint spacing
d	Minimum significant accident delay duration
E_f	Set of vulnerability factors
e_i	Geological unit with discretization number i
e_i^n	Neighboring units surrounding e_i
e_m^v	Vulnerability-related factors with total number of m
f_ad	Probability distributions for accident recovery
f_ak	Foundation bearing capacity
f_rk	Uniaxial saturated compressive strength
f_sd	Probability distributions for step durations
g_v	Level of tunnel vulnerability
H_i	Information entropy value
$\boldsymbol{J}$	Judgment matrix of AHP with internal values a_{ij}
J_s	Spatial correlation parameter
K_u	Kurtosis of probability distribution functions
K_v	Rock mass integrity coefficient
L	Set of all lithology categories within the study area
M	Median of probability distribution functions
M_i	Centrality of evaluation factors
N_c	Total number of excavation cycles
N_i	Local neighborhood system surrounding e_i
N_j	Number of joint sets
N_s	Total number of MC simulations
n_c	Current excavation cycle number
n_s	Current simulation iteration
p_j	Probability of each label for a unit
q_sa	Pile side soil friction resistance
R_c	Uniaxial saturated compressive strength
R_k	Geological configuration generated in the kth iteration
$\bar{R}$	The most possible configuration
r_i	Lithology category value of e_i
r_{N_i}	Lithology category values of neighborhood units
S	Set of geological units
S_k	Skewness of probability distribution functions
$\boldsymbol{T}$	Comprehensive influence matrix with values t_{ij}
T_e	System temperature
T_st	Tunnel face stability time
t_a	Durations of accident recovery
t_s	Durations of construction steps
v_rs	Value of random seed
W	Comprehensive calculated weight of factors
Z	Normalized partition function
$\mathfrak{R}$	Configuration space composed of all R_k
λ	Rate parameter of the exponential distribution
μ	Mean of probability distribution functions
ρ	Natural density of the stratum
σ	Standard deviation of probability distribution functions
φ	Internal friction angle of the stratum
φ_i	Direction representing the dip angle of the stratum
φ_int	Estimated dip angle values
θ_in	Differences in polar angles between e_i and e_i^n
λ_max	Maximum eigenvalue of $\boldsymbol{J}$

efficiency and ensuring robust safety management in tunnel projects.

1.2 Related works and problem statement

From a theoretical perspective, the repetitive nature of tunnel excavation cycles positions process simulation modelling (PSM) as an effective approach for analyzing construction uncertainties (Rahm et al., 2016), particularly in optimizing resource allocation and construction sequencing (AbouRizk, 2010). Vick and Einstein (1974) were among the first to apply PSM in tunnel engineering. Since then, advanced PSM tools have been developed, including CYCLONE-CPM (Bi et al., 2015), SD-DES (Dang et al., 2018), and CSM2020 (Khetwal et al., 2022). Recent developments emphasize refining simulation techniques and integrating complementary analytical tools to better align with practical engineering demands, as outlined in Table 1. Among these methods, discrete event simulation (DES) stands out for its ability to model and analyze the complexity and interdependence of events or processes, offering greater accuracy and flexibility (Abdelmegid et al., 2020). DES simulates events at discrete time points, triggering changes in the system's state (Sharma, 2015). When combined with Monte Carlo (MC) simulation, which generates random variables for stochastic analysis, DES effectively addresses both probabilistic and deterministic problems (Metropolis & Ulam, 1949).

Sousa and Einstein (2021) emphasized that, aside from human management errors, unforeseen geological conditions constitute the primary cause of tunnelling accidents. Geological conditions have also been integrated into several PSM models. The decision aids for tunneling (DAT) employ discrete information points along tunnel alignments to dynamically update geological conditions, thus optimizing resource allocation and planning processes

Table 1
Examples of improved PSM models in tunnel engineering.

Authors (year)	PSM method	Method for improvement	Improved aspects	Application
Khetwal et al. (2022)	DES	Optimized CSM2020 model	Enhanced estimation of machine utilization	TBM tunnels
A. Khetwal et al. (2021a)	DES	6 empirical comparison models	Model comparison and validation	TBM tunnels
S. Khetwal et al. (2021b)	Monte Carlo	Stochastic event simulation model	Assessment of functionality loss	Road tunnels
Frough et al. (2019)	DES	Double shield tunnel database	Prediction of TBM performance	TBM tunnels
Dang et al. (2018)	DES	System dynamics (SD)	Evaluation of soil composition impact	Micro-tunnelling
Kim et al. (2018)	WebCYCLONE	Convolutional Neural Network	Cost reduction in earthmoving process	NATM tunnels
Yu et al. (2017)	CYCLONE	Bayesian-based risk evaluation model	Mitigation of risk-caused delay uncertainties	Diversion tunnels
Rahm et al. (2016)	Process simulation	Extended disturbance simulation	Analysis of technical failure and supply chain-caused delays	Mechanized Tunnelling
Bi et al. (2015)	CYCLONE	CYCLONE-CPM Simulation structure	Real-time construction schedule analysis	Diversion tunnels
Al-Bataineh et al. (2013)	Simphony.NET	Scenario-based simulation analysis	Consideration of variable breadth and scope	Utility tunnels

(Haas & Einstein, 2002; Min & Einstein, 2016). Additionally, Yu et al. (2017) integrated Bayesian networks within the CYCLONE model to systematically incorporate conventional risks and risk events, enabling detailed operational-level risk assessments. Nevertheless, existing studies on geological uncertainty modelling predominantly focus on one-dimensional analyses along the tunnel axis, lacking a comprehensive investigation into spatial characteristics such as variability and anisotropy of stratum distribution.

The modelling of geological uncertainty is principally divided into two main research directions (Wang et al., 2018b). The first employs geostatistical methods like the Kriging method, sequential Gaussian models, and multiple-point geostatistics to describe variables regionally through variogram functions (Bian et al., 2024). However, these models often fall short in capturing the complex cyclicity and environmental interplays in lithological sequences (Bi et al., 2015). The second approach utilizes Markov models, such as Markov random field (MRF) and chain models (Wang et al., 2024a), to construct stochastic geological profiles. This method has been applied in the assessment of slope and tunnel stability (Wang et al., 2016; Wang et al., 2018b). MRF methods, aligning closely with strata formation processes and spatial heterogeneity, provide more interpretable geological insights compared to variogram-based approaches (Elfeki & Dekking, 2001; Wang, 2020). Additionally, by discretizing geological profiles using MRF, the resulting units align with the cyclical process of DES simulation, thereby enhancing the integration of model modules. Consequently, analyzing geological uncertainty via MRF emerges as a logical progression towards achieving more accurate simulation results (Cho, 2007; Fenton & Griffiths, 2008).

With the rapid advancement of internet-based information technology, new possibilities have emerged for innovating tunnel PSM models (Phoon et al., 2022). Current PSM models largely remain theoretical and disconnected from on-site realities (Persson & Araldi, 2009), leading to slow, inaccurate responses to site conditions and limiting their ability to provide actionable guidance to stakeholders, thus reducing their practical value (Yu et al., 2023). The core issue lies in the lack of effective interaction and deep integration between construction sites and digital models (Li et al., 2024a). In the context of transitioning towards Industry 4.0 (Phoon et al., 2024), the introduction of digital twin (DT) technology offers a potential solution.

The contemporary consensus defines DT as a theoretical five-dimensional framework comprising physical entities, virtual entities, connections, DT data, and services (Liu et al., 2024; Tao et al., 2018). DT describes, maps, and predicts the semantic, geometric, and behavioral characteristics of physical entities through virtual representations and data-driven models (Samaniego et al., 2020), thereby enabling functions such as real-time perception, dynamic control, and information sharing (Tao et al., 2019b). Since its introduction to tunnel engineering around 2020, DT has been the focus of recent studies exploring its application (Opoku et al., 2021). Ye et al. (2023) developed a DT-based early warning and safety management platform, simulating tunnel construction and geological scenarios. Fang et al. (2024) developed a state digital twin model that integrates building information models and point cloud data to assess over-and under- excavation during the tunnel construction process. Shen et al. (2022) devised a DT-based design methodology for tunnel interior lighting and decoration, merging virtual reality (VR), numerical simulations, and field experiments.

These studies highlight DT's potential to integrate physical construction sites into the virtual domain, enabling cost-effective lifecycle simulation and optimized construction (Li et al., 2024a). However, the application of DT in tunnel PSM modelling remains largely unexplored, with significant challenges in translating DT from theory to practice, including integrating diverse data sources, ensuring data interoperability in complex underground environments, and developing systems for dynamic analysis and

feedback services. Additionally, its practical effectiveness requires further investigation.

1.3 Objectives

To address the research gap, the specific objectives of this study are as follows: (i) Propose a PSM model capable of considering multi-source uncertainties in NATM tunnel construction; (ii) Develop a DT framework to provide timely and effective guidance for tunnel construction; (iii) Explore the application effects of an intelligent construction system integrating the DT concept and PSM in a real-world case.

Therefore, an innovative DT-enabled simulation system for NATM tunnel construction considering multi-source uncertainty is developed. The main contribution of this paper is as follows: (i) An integrated PSM model was developed to structurally simulate NATM tunnel construction, capturing its components and construction processes; (ii) Modules were developed for the PSM to enable the evaluation of geological and construction-related uncertainties, including those related to conventional delays and accidents; (iii) A five-dimensional DT framework for tunnel construction guidance was proposed, integrating perception, transmission, modelling, and service layers. The research findings have been validated in a NATM tunnel project in Yunnan Province, China, demonstrating significant savings in both construction time and cost.

The remaining structure of the paper is as follows. Section 2 describes the specific methods employed for system construction and model development. A case study of an actual tunnel construction project is presented in Section 3. Finally, Section 4 outlines the conclusions and future work.

2 Methodology

The methodology for developing the DT-enabled simulation system comprises two main components: the establishment of a system framework and the construction of an integrated PSM model, as illustrated in Fig. 1. Drawing upon the concept of the 5D-DT, the system framework is structured into four interconnected layers: the perception layer, transmission layer, modelling layer, and the alternation layer. To systematically address uncertainties inherent in the construction process, the modelling layer incorporates the integrated PSM model, which serves as the analytical core of the digital domain. This model consists of three interrelated modules. Module I addresses geological uncertainty by integrating borehole data with prior geological knowledge and employs MRF to simulate geologic distributions. Module II evaluates the risk of construction accidents by integrating geologic data with engineering measurements and expert judgments, utilizing the analytic hierarchy process – decision-making trial and evaluation laboratory (AHP–DEMATEL) method to assess accident vulnerability levels. Module III synthesizes the uncertain-

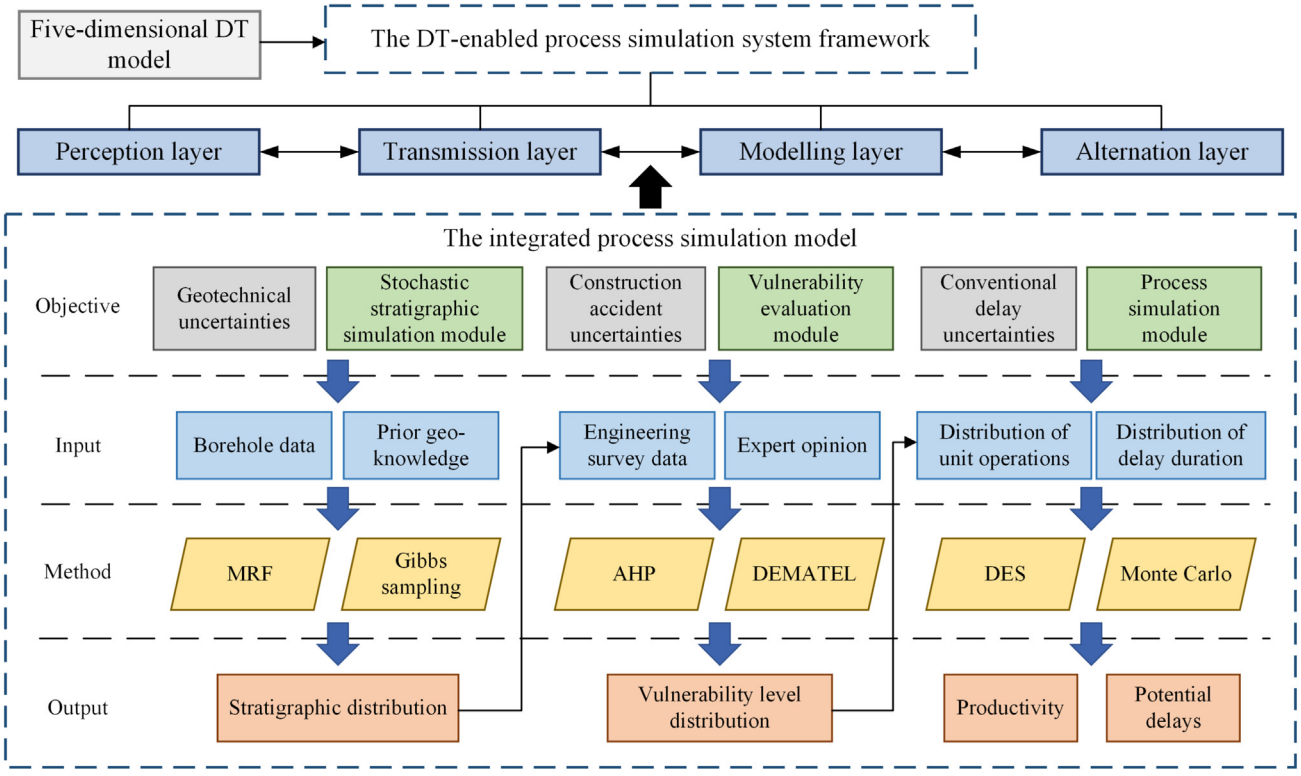


Fig. 1. Methodological roadmap for developing the DT-enabled simulation system.

ties from geological conditions and construction accidents with conventional delay factors, using the DES and MC simulation to forecast construction productivity and potential delays.

2.1 Digital twin framework

Based on the concept of the five-dimensional DT model M_{DT} = (PE, VE, Ss, DD, CN) (Tao et al., 2019a), which corresponds to physical entities (PE), virtual entities (VE), services (Ss), twin data (DD), and connections (CN), we developed a 5D DT-enabled process simulation system framework tailored for guiding NATM construction, as illustrated in Fig. 2. At the implementation level, the framework comprises the perception layer (sensing PE and collecting DD), the transmission layer (facilitating bi-directional CN), the modelling layer (constructing VE), and the alternation layer (offering Ss and providing CN from VE to PE), as depicted in Fig. 3.

The perception layer serves as the foundational element of the system architecture, designed to efficiently collect DD from onsite PE (e. g., workers, construction machinery, tunnel structures, and geology). To enable comprehensive data collection, an array of methods has been implemented on-site, including wearable data loggers, machinery parameter data collection systems, mobile data collection vehicles, multifunctional sensors, and video surveillance. Additionally, the perception layer integrates relevant documentation from both pre-construction and construction stages. To manage these heterogeneous and multi-source data, the system incorporates a configurable data access service framework, supporting modular expansion and multiple data access methods such as SQL Server, NetMQ, SignalR, WebAPI, Excel, and device protocols.

The transmission layer, representing the CN dimension of DT, enables seamless integration across its components. It efficiently relays data associated with geographic locations (GIS coordinates), engineering sections (3D identifiers), and database indices. The layer utilizes a robust wireless network composed of multimodal self-organizing nodes, which are made up of access points (APs), directional antennas, and wireless bridges, powered by battery packs. Nodes placed every kilometer enable real-time transmission of construction data to a base station at the tunnel entrance, centralizing the data for the control center. This setup enables real-time monitoring of tunnel safety and progress while connecting site personnel and machinery to the Internet for immediate access to analytical results and system alerts.

The modelling layer creates a corresponding VE based on PE and DD. This layer integrates structured data, 2D drawings, and 3D models using graphic engine technologies such

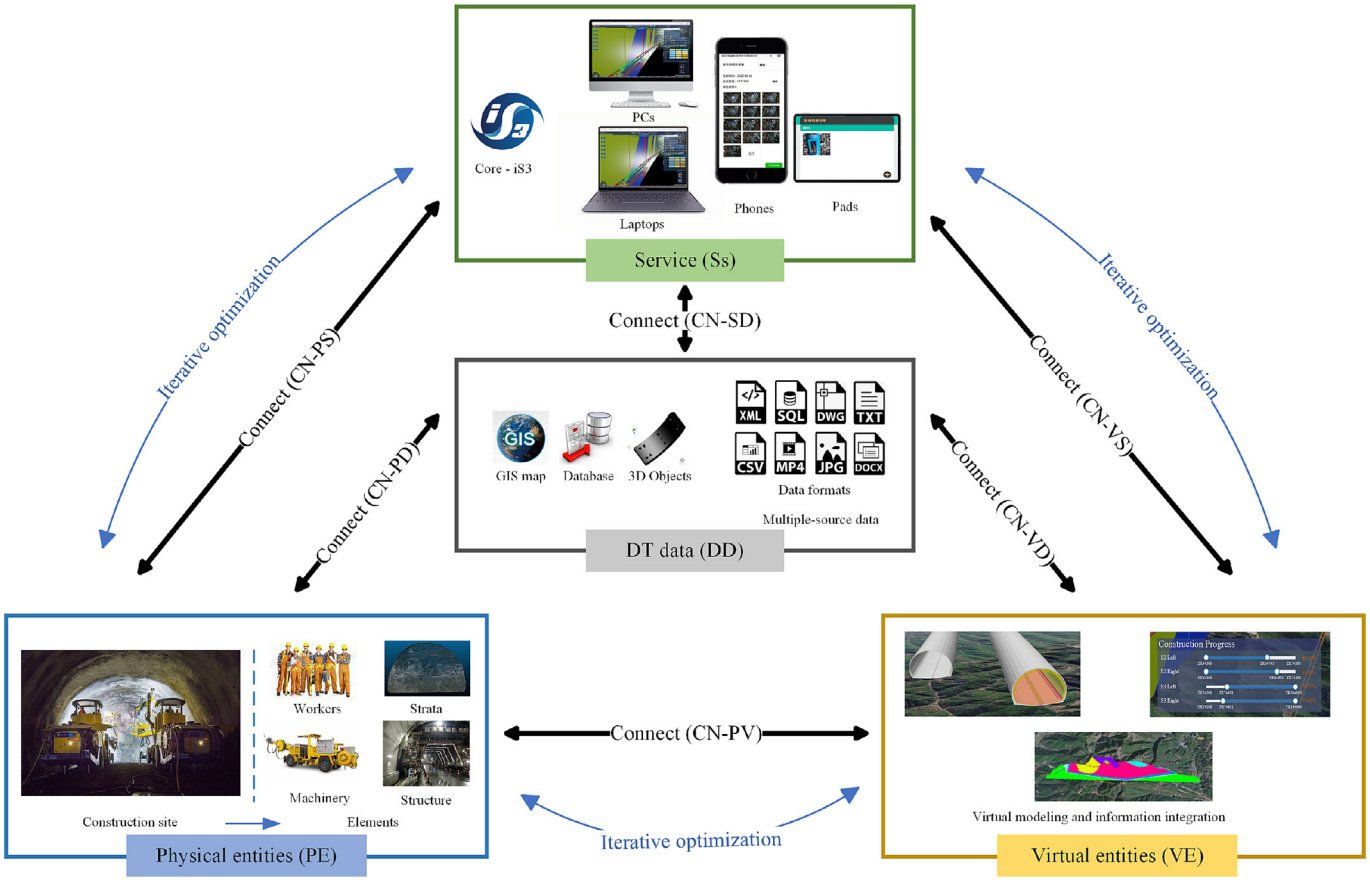


Fig. 2. Five-dimensional DT conceptual model.

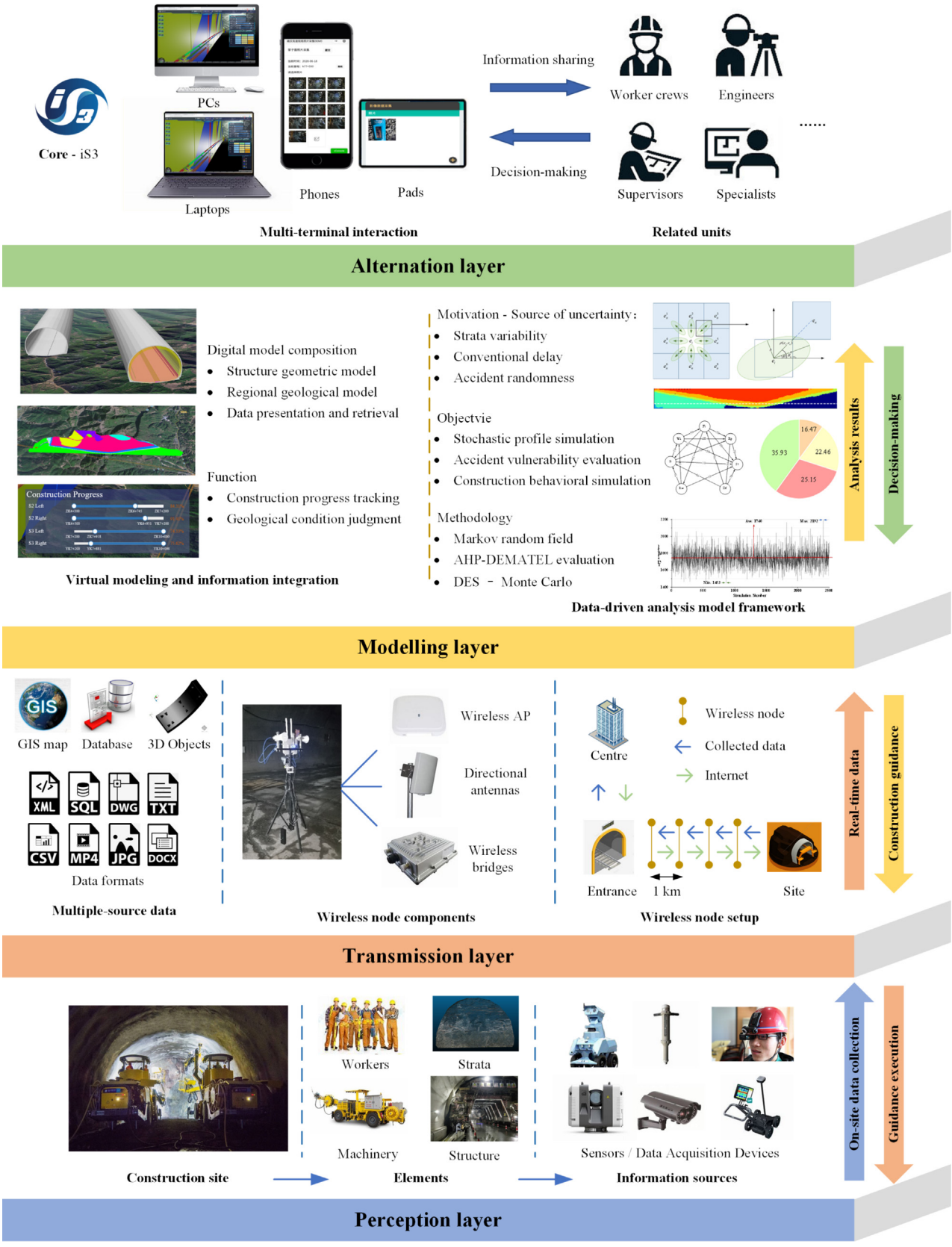


Fig. 3. System layered architecture.

as building information modeling (BIM), geographic information system (GIS), and OpenGL. Resources are organized into digital objects to create detailed virtual models of tunnel structures and regional geology, placed within a virtual environment mirroring the project site. By linking virtual entities to databases containing survey, design, and progress information, the system enables dynamic mapping between physical and virtual entities. This facilitates real-time visualization and efficient retrieval of historical data, serving as a core tool for dynamic project management.

The uppermost layer, the alternation layer, delivers Ss to guide tunnel construction. Serving as a directed CN from VE to PE, it ensures real-time feedback of analytical results and supports on-site collaborative operations. Based on the infrastructure smart service system (iS3) framework (Zhu et al., 2018) and cloud technology, it offers interactive services via desktop (web) interfaces and mobile applications, ensuring compatibility with devices like laptops, desktops, phones, and tablets. It enables the immediate communication of data-driven model outcomes, such as geological analysis, construction progress, and risk assessments, to stakeholders, including workers, engineers, managers, and experts. By facilitating dynamic interactions and informed decision-making for onsite personnel, it also provides access to tailored historical data, streamlining project management.

2.2 Process simulation model development

2.2.1 Structured analysis and modelling

In tunnel and linear infrastructure construction, engineering tasks often adhere to a sequential, repetitive pattern, establishing a predefined operational sequence (Ökmen & Öztaş, 2008). To enhance the accuracy and depth of the PSM model for NATM construction, we adopted the systems modeling language (SysML) (Friedenthal et al., 2014). This innovative language employs a variety of diagram types to depict system requirements, physical constraints, structures, and behaviors (Dang et al., 2018). Specifically, we used the block definition diagram (BDD) to detail the construction site's structural elements and fundamental units, termed "blocks." Additionally, the sequence diagram (SQ) illustrated the dynamic interactions and operational logic among these blocks, as depicted in Figs. 4 and 5.

In the BDD depicted in Fig. 4, each on-site PE is defined as a unit (block) (Dang et al., 2018). The construction site primarily comprises four fundamental units: personnel, machinery, structure, and geology. The collaborative execution system integrates specialized personnel and various equipment. In the Labour block, the crews are divided based on tasks, although on-site personnel may take on multiple tasks. As depicted in Fig. 5, the execution logic of NATM construction unfolds in the following sequence: (i) Initiation stage: The construction cycle begins, and workers utilize horizontal drilling machines or geophysical methods to assess the geological conditions ahead. In cases of loose or fractured strata, advanced support structures are implemented. (ii) Drilling and blasting: Under the engineers' drilling and charging directives, the drilling crew operates drilling jumbos for borehole creation, followed by the blasting crew and explosive loading trucks executing charging and blasting. (iii) Safety and clearance: Post-blasting, a safety inspection interval is observed, after which the excavation team employs shovels and conveyors for debris removal. (iv) Primary support: The lining crew applies initial shotcrete via wet shotcrete machines and erects preliminary support structures, including steel frames, anchor bolts, and steel mesh. (v) Secondary lining: Utilizing the lining trolley, the team lays the waterproof layer and performs secondary lining of the arch wall. Following each cycle, engineers assess the outcomes and initi-

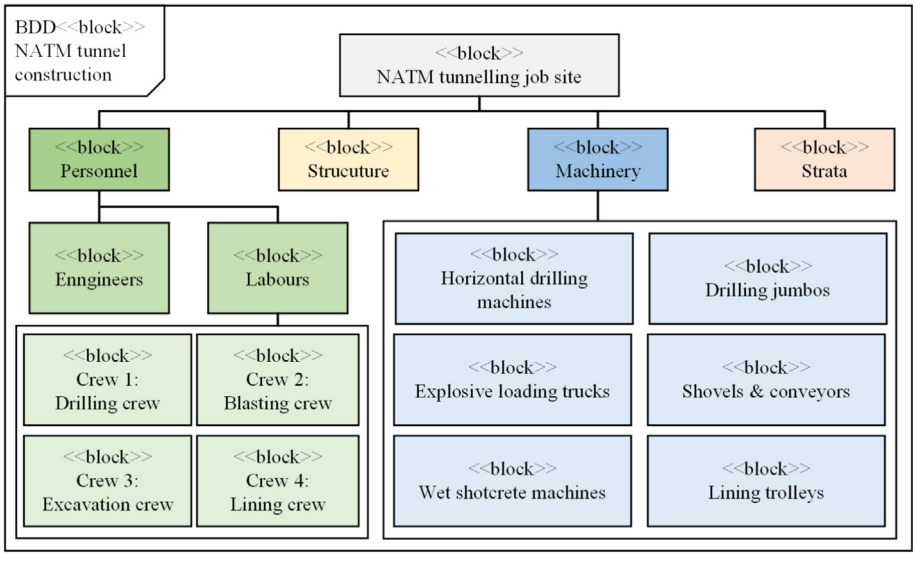


Fig. 4. Block definition diagram for NATM tunnel construction site.

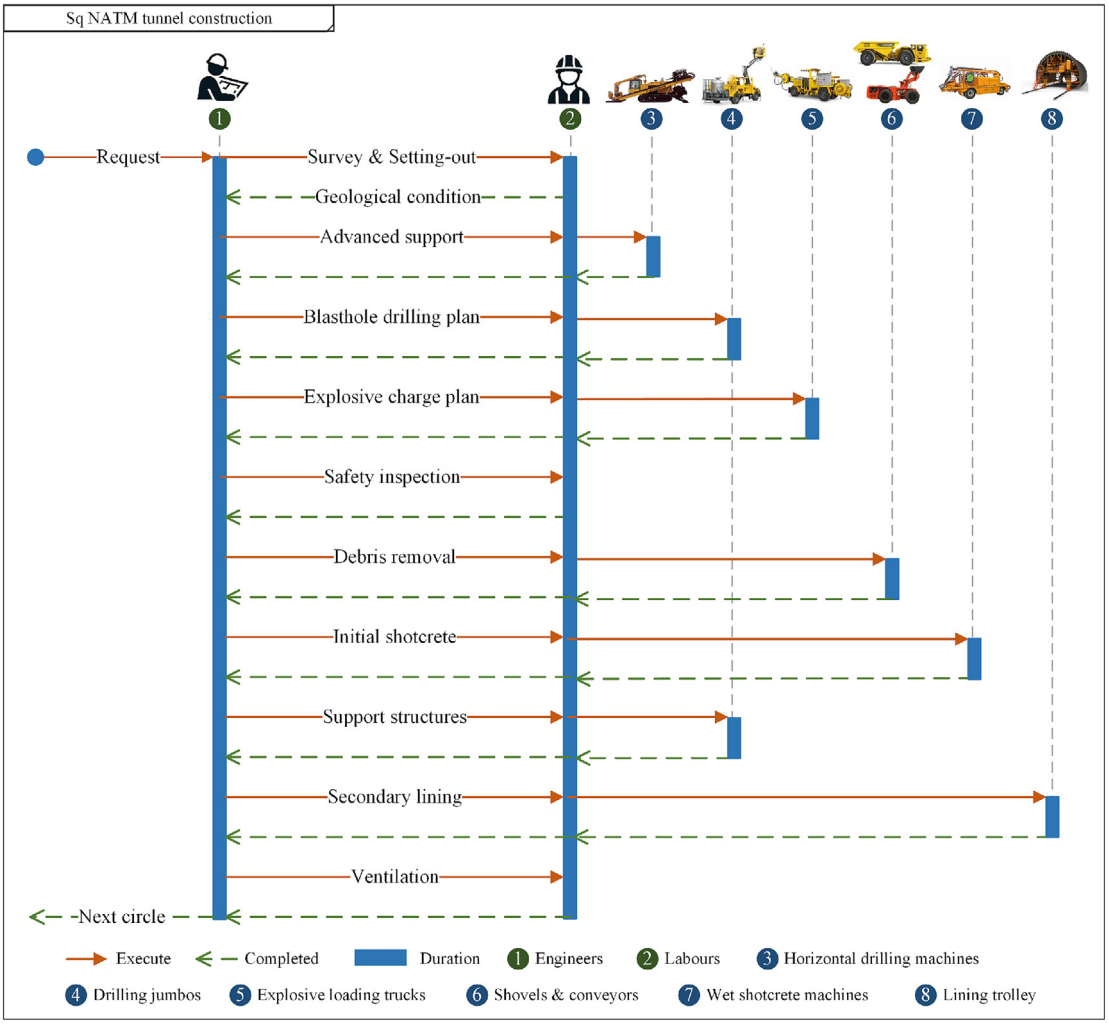


Fig. 5. Sequence diagram of the NATM tunnelling process.

ate the subsequent cycle, a process that repeats until tunnel construction completion.

Based on the structured analysis, we chose the AnyLogic simulation software (version 8.8.2) to develop and execute the DES model, as illustrated in Fig. 6. AnyLogic stands out for its versatility and integration of diverse simulation methodologies within a Java-based platform, making it highly suitable for process modelling and probabilistic evaluations (Borshchev, 2014). This model comprises various components and centers around the main agent, which manages data exchange, event scheduling, and outputs. Key elements to the model are AnyLogic's process modelling library blocks, like "Source" for initiating construction cycle simulations, "Move To" for process progression, and "Service" for simulating specific construction operations by allocating resources, calculating operation duration, and releasing resources for subsequent operations. Additionally, the model employs variables and parameters to monitor essential metrics like time and resource utilization, and function bodies for logic processing and on-site data integration. Results are systematically stored in spreadsheets and databases for subsequent analysis. The model's components were meticulously coded, interconnected, and verified to ensure error-free operation.

2.2.2 Double-loop process simulation

Leveraging the DES model, we formulated a simulation execution strategy, outlined in Fig. 7. Reflecting the NATM tunnels' sequential and cyclical excavation process, we established three distinct simulation clocks: a global model clock (clock-g), a single simulation clock (clock-s), and an excavation cycle clock (clock-c). These clocks utilize a next-event time advancement mechanism to propel the simulation. This setup allows for precise monitoring and documentation of simulation activities at the global level, for individual simulations, and for each specific excavation cycle. The methodology unfolds as follows:

(1) Outer loop (MC Simulation): The outer loop consists of N_s iterations of the complete tunnel construction MC simulation. Firstly, the model is initialized with

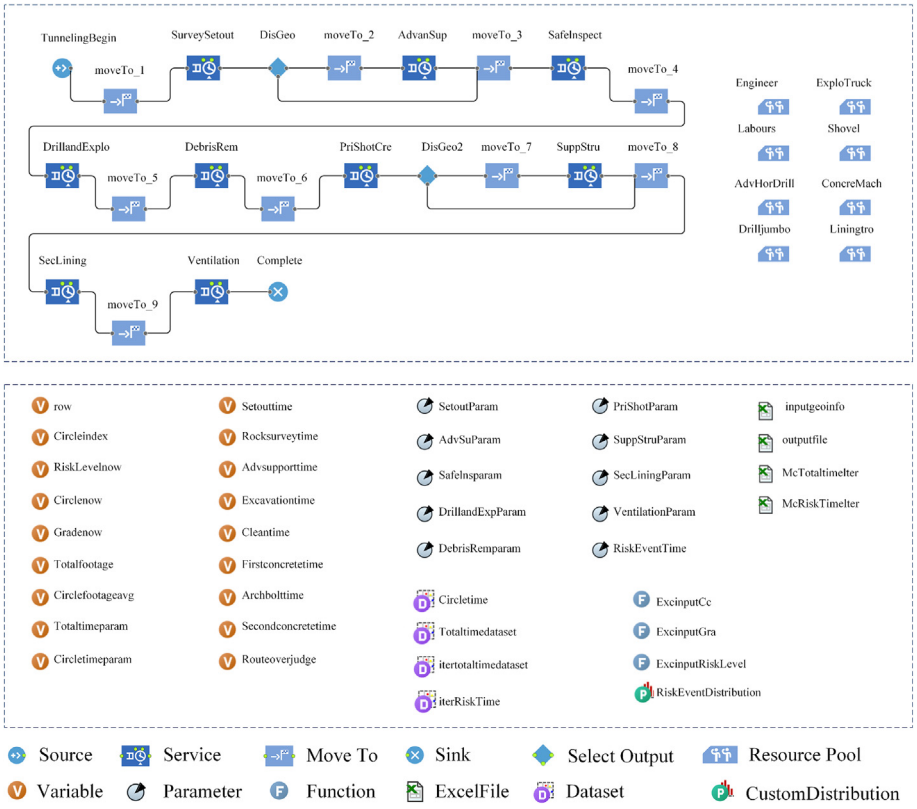


Fig. 6. Discrete event process and model components.

built-in parameters such as the number of simulations N_s, value of random seed v_{rs}, and probability distributions for the durations of regular construction steps f_{sd} and accident recovery time f_{ad}, etc. The clock-g is then initialized and starts timing, and the first iteration begins ($n_s = 1$). During each iteration, the clock-s records the time taken, as details provided in the inner loop. After each simulation, it checks whether the current iteration n_s has reached N_s. If not, it proceeds to the next iteration; otherwise, the MC simulation concludes and records the time on the clock-g.

(2) Inner loop (DES simulation): A single MC simulation consists of consecutive excavation cycles, each comprising a series of construction steps. The loop encompasses these critical actions:

(i) At the start of the inner loop, the excavation cycle count n_c is set to 1, and the clock-c is initialized and starts timing. The geological condition of the tunnel section for the current excavation cycle is determined by the geological uncertainty analysis module based on MRF.

(ii) Based on the geological conditions and other relevant engineering information, the tunnel vulnerability level g_v at the current section is assessed via the AHP–DEMATEL method. This assessment determines the likelihood of accidents and the critical path of the construction steps. Subsequently, the durations of sequential steps t_s and accident recovery t_a are computed using their respective probability distributions f_{sd} / f_{ad}.

(iii) The calculated t_a and t_s determine the timing for the clock-c. The loop then checks if n_c has met the total number of excavation cycles N_c, which is determined by whether the cumulative advance has reached the tunnel length. If not, it proceeds to the next cycle; if so, the simulation round concludes, and control returns to the outer loop.

2.3 Stochastic geological profile

2.3.1 Random field construction

Reliable geological information plays a crucial role in NATM tunnel construction and simulation (Wang et al., 2024b). Therefore, this study incorporates geological uncertainty through an MRF model (Wang et al., 2018a), as illustrated in Fig. 8. The two-dimensional geological profile is discretized into a uniform grid of square units $S = \{e_i | i = 1, 2, \cdots, n\}$, with i denoting the unit index. Each unit e_i is surrounded by eight neighboring units forming its local neighborhood system $N_i = \{e_i^{n1}, e_i^{n2}, \cdots, e_i^{n8}\}$. Spatial correlations between units capture the variability and dependencies of geological properties.

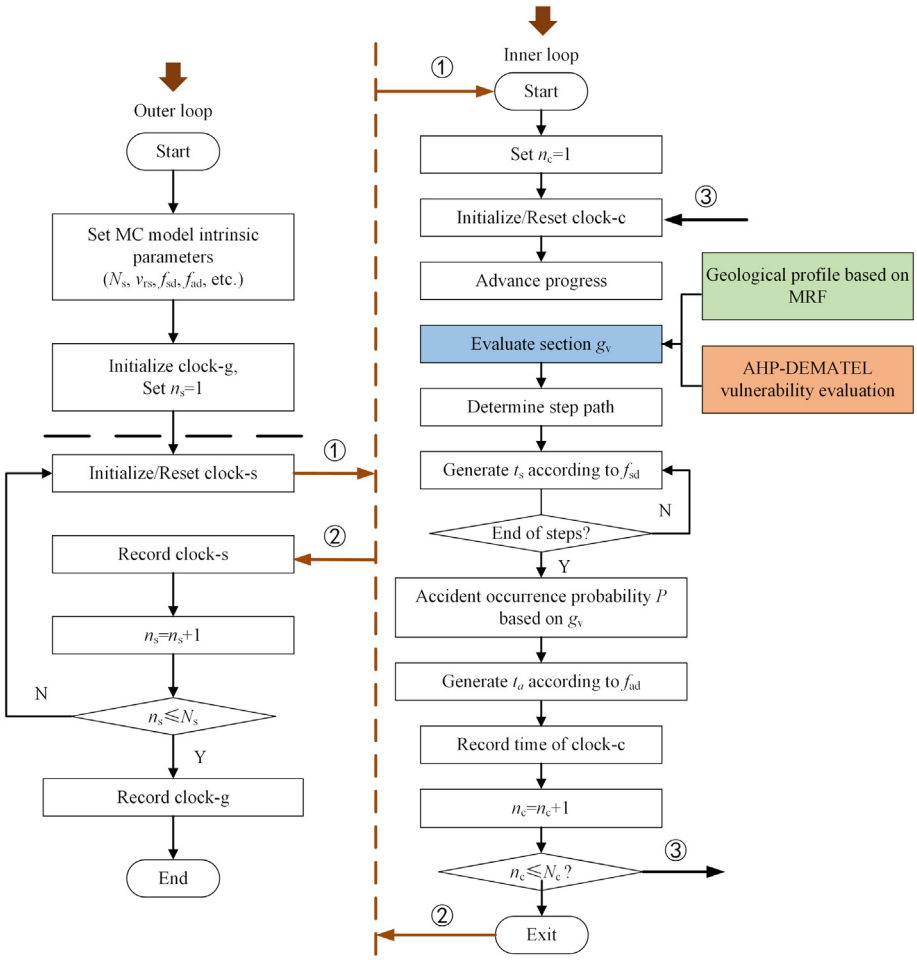


Fig. 7. Execution framework of the dual-loop simulation strategy.

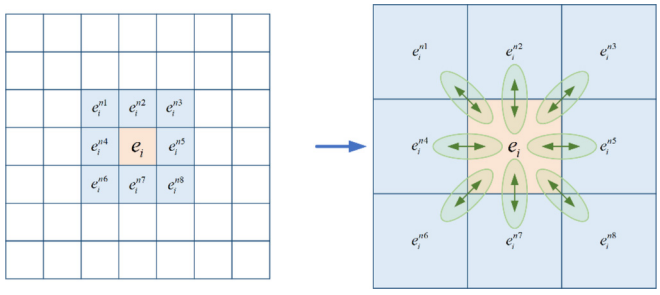


Fig. 8. Unit discretization and neighborhood system.

An initial lithology category r_i is randomly assigned to e_i from the set L, which includes all lithology categories in the model domain. This initial assignment serves as the starting point for generating a geological configuration $R_k = \{r_i | r_i \in L, e_i \in S\}$. Through iterative k updates, spatial dependencies and correlations are progressively incorporated, resulting in a configuration that most likely represents geological plausibility. The set of all such configurations constitutes the configuration space $\Re = \{R = R_k | r_i \in L, e_i \in S\}$. A configuration R_k satisfies the Markov property, as defined by Eqs. (1) and (2), ensuring each unit's state depends only on its local neighborhood.

$$P(r_i | r_{S-N_i}) = P(r_i | r_{N_i}), \tag{1}$$

$$P(R = R_k) > 0, \ R_k \in \Re. \tag{2}$$

Subsequently, a search is conducted to identify the configuration $\bar{R}$ that maximizes the joint probability density $P(R)$, representing the most likely geological configuration, as shown in Eq. (3). However, the local nature of the Markov property, as defined in Eq. (1), makes direct computation of $P(R)$ challenging. To address this, the Hammersley–Clifford theorem (Clifford, 1990) establishes an equivalence between MRFs and Gibbs distributions, allowing $P(R)$ to be expressed using the Gibbs distribution measure $\mu(R)$, as shown in Eq. (4).

$$\bar{R} = \arg\max_{R \in \Re} P(R), \tag{3}$$

$$P(R) = \mu(R) = \frac{1}{Z} \exp[-U(R)/T_e], \tag{4}$$

where $U(R)$ represents the energy function corresponding to state R. The variable T_e denotes the system temperature, controlling the scaling of energy. Z symbolizes the normalized partition function, as illustrated in Eq. (5).

$$Z = \sum_{R \in \Re} \exp\left[-U(R)/T_e\right]. \tag{5}$$

2.3.2 Gibbs sampling solution

To solve the maximization problem, we employed the efficient Gibbs sampling method. The method iteratively updates r_i based on the latest lithology category values of neighborhood units r_{N_i} through $P(r_i|r_{N_i})$, thereby transitioning from R_k to the next configuration R_{k+1}, and evaluates the current $U(R)$. $U(R)$ is calculated by summing the potential function values $V_{c_i}(r_i)$ of potential groups c_i formed by (r_i, r_{N_i}), as shown in Eq. (6), where C represents the set of c_i.

$$U(R) = \sum_{c_i \in C, r_i \in R} V_{c_i}(r_i, r_{N_i}), \tag{6}$$

where $V_{c_i}(r_i, r_{N_i})$ reflects the spatial correlation between r_i and r_{N_i}. To enhance the efficiency of processing operations for large-scale units, we selected a structurally concise pair of cliques and employed the Potts model (Wu, 1982) to express the potential function, as illustrated in Eq. (7).

$$V_{c_i}(r_i, r_{N_i}) = -J_s \sum_{(e_i, e_n)} \delta(r_i, r_{N_i}), \tag{7}$$

where J_s is a spatial correlation parameter, which represents a constant characterizing the strength of interaction between units of the same lithology. $\delta(r_i, r_{N_i})$ is the Kronecker delta function (Brewer, 1978), utilized to ascertain whether two adjacent units possess the same lithology, as demonstrated in Eq. (8).

$$\delta(r_i, r_{N_i}) = \begin{cases} \rho\left(e_i, e_i^n\right) & \text{if} \quad r_i = r_{N_i} \\ 0 & \text{if} \quad r_i \neq r_{N_i} \end{cases}, \tag{8}$$

where $\rho\left(e_i, e_i^n\right)$ represents a function that captures the spatial correlation of (r_i, r_{N_i}). To this end, a reference ellipse is constructed for each pair $\left(e_i, e_i^n\right)$, as shown in Fig. 9, set according to the global polar coordinate system, with its major axis orientation φ_i determined by the existing information on stratum dip angle. The ellipse's minor axis length b is fixed at 1.0, while the major axis length a is defined as an intensity parameter measuring spatial correlation. $\rho\left(e_i, e_i^n\right)$ is set to the semi-chord length in the direction from e_i to e_i^n, calculated as Eq. (9), reflecting two factors of geologic spatial correlation: (i) Anisotropy, i.e., the extension of strata is usually direction-dependent, indicated by the major axis direction φ_i to reflect the tangential direction of structural interfaces; and (ii) Variability, i.e., geologic properties differ at different locations, represented by the differing polar angles θ_{in}.

$$\rho\left(e_i, e_i^n\right) = ab \Big/ \sqrt{\left[b\cos\left(\theta_{in} - \varphi_i\right)\right]^2 + \left[a\sin\left(\theta_{in} - \varphi_i\right)\right]^2}. \tag{9}$$

2.4 Uncertain delays analysis

As described in Section 1, the uncertainties in the construction process mainly include conventional delay factors and construction accidents. The conventional delay factors can be described through continuous probability distributions. In contrast, the causes and impacts of construction accidents display greater complexity and uncertainty. Therefore, we focused on exploring key factors affecting tunnel vulnerability, such as hydrogeological conditions, rock properties, and rock mass structure. We employed the AHP–DEMATEL method to model the interactions among these factors, assessing tunnel vulnerability in each excavation cycle. Based on this assessment, we further analyze the probability of construction accidents and the potential delays they may cause.

2.4.1 Tunnel vulnerability evaluation

The analytic hierarchy process (AHP) hierarchically structures multi-criteria evaluation problems by assigning relative importance weights to factors, but it overlooks interdependencies (Al-Harbi, 2001). In contrast, DEMATEL applies graph theory and matrix analysis to examine factor interrelations (Si et al., 2018). We integrated the AHP and DEMATEL methods to comprehensively evaluate factor importance and interdependencies, improving

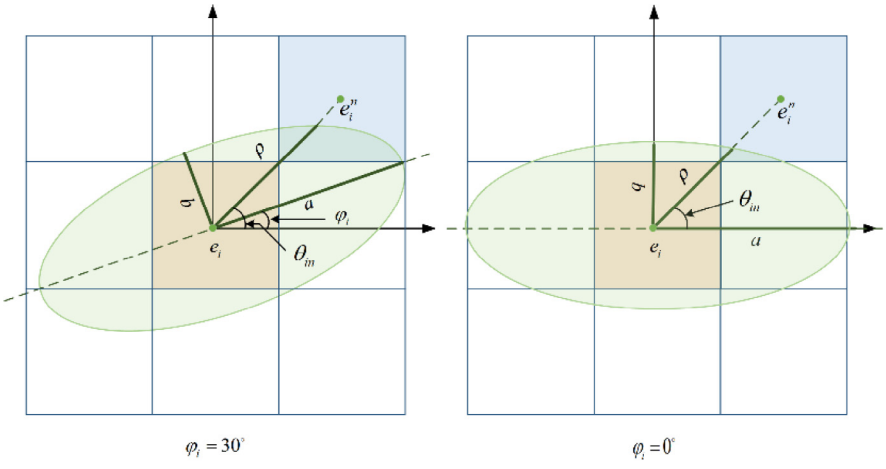


Fig. 9. Reference ellipse for the unit pair.

objectivity and precision, as illustrated in Fig. 10. The methodology involves the following steps:

Step 1: Gather opinions and select factors: Gather expert opinions to determine the relevant factor set $E_f = (e_1^v, e_2^v, \cdots, e_m^v)$, where m is the total number of factors.

Step 2: Establish relationships with AHP–DEMATEL: Utilize AHP to establish hierarchical relationships among factors and apply DEMATEL to evaluate their direct impacts. Visualize these interdependencies in a causal relationship diagram with weighted directed edges representing the strength and direction of influence.

Step 3: Construct the judgment matrix: Perform pairwise comparisons to build a judgment matrix $\boldsymbol{J} = (a_{ij})_{m\times n}$. To conduct a consistency check, calculate the maximum eigenvalue λ_{max}, consistency index CI, and consistency ratio CR, as Eqs. (10) and (11). Here, RI represents the random consistency index, which is related to the matrix order. If CR < 0.1. the consistency of the judgment matrix $\boldsymbol{J}$ is considered acceptable. Then, the relative importance weights W^1 are synthesized using the eigenvalue method, as indicated in Eq. (12).

$$\mathrm{CI} = \frac{(\lambda_{max} - n)}{n-1}, \tag{10}$$

$$\mathrm{CR} = \mathrm{CI}/\mathrm{RI}, \tag{11}$$

$$W^1 = (\omega_1^1, \omega_2^1, \cdots, \omega_n^1) = \sqrt[n]{\left(\prod_{j=1}^{n} a_{ij}\right)}. \tag{12}$$

Step 4: Build the influence matrix: Using DEMATEL, construct a direct influence matrix $\boldsymbol{A} = (b_{ij})_{n\times n}$, normalize it to $\boldsymbol{N}$, and derive the comprehensive influence matrix $\boldsymbol{T} = (t_{ij})_{n\times n}$, as described in Eqs. (13)–(15). Next, calculate the influence degree D_i, being-influenced degree B_i, and the centrality M_i. Finally, normalize the centrality to determine the influence weight W^2 of each element, as outlined in Eqs. (16)–(19).

$$m = \max\left(\sum_{j=1}^{n} a_{ij}\right), \tag{13}$$

$$\boldsymbol{N} = \left(\frac{a_{ij}}{m}\right)_{n\times n}, \tag{14}$$

$$\boldsymbol{T} = \boldsymbol{N}(\boldsymbol{I} - \boldsymbol{N})^{-1}, \tag{15}$$

$$D_i = \sum_{j=1}^{n} t_{ij}, \tag{16}$$

$$B_i = \sum_{j=1}^{n} t_{ji}, \tag{17}$$

$$M_i = D_i + B_i, \tag{18}$$

$$W^2 = (\omega_1^2, \omega_2^2, \cdots, \omega_n^2) = \frac{M_i}{\sum_i^n M_i}. \tag{19}$$

Step 5: Synthesize and normalize weight vector: Combine the results from AHP and DEMATEL according to the specified formula $W = W^1 W^2$, and normalize it to obtain standardized weights $\bar{W}$.

In examining the prevalent causes of construction accidents, alongside the types of data available from construction cases, we identified critical factors E_f for accident vulnerability assessment. These encompass hydrogeological conditions, rock physical and chemical stability, and rock mass structural properties. Consequently, we selected the hydrogeological richness (Rw), seepage status (Sp), weathering intensity (Wi), fractured state (Fs), joints and fissures development (Df), surrounding rock stability (St), and surrounding rock property (Rp) as primary assessment factors for tunnel construction accident vulnerability.

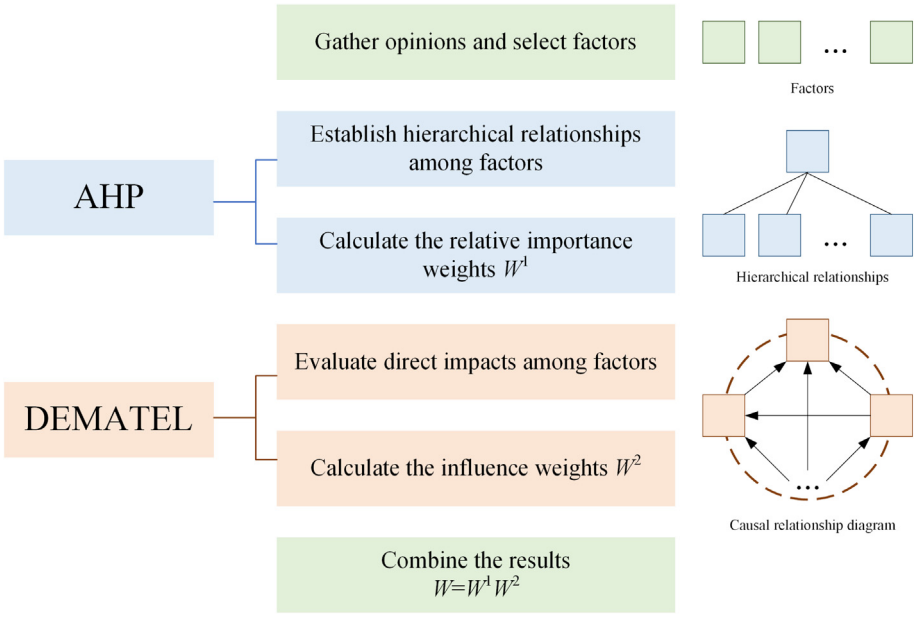


Fig. 10. Framework of the AHP–DEMATEL method.

To perform quantitative grading calculations, the key parameters for each evaluation factor were determined, including maximum borehole water inflow rate Q_{max}; seepage volume per 10 m of tunnel length per minute Q_s; weathering coefficient K_f; rock mass integrity coefficient K_v; number of joint sets N_j; joint spacing D_j; tunnel face stability time T_{st}; and uniaxial saturated compressive strength R_c. Based on these parameters, a comprehensive grading and classification system was developed to assess vulnerability, ranging from high (level I) to low (level IV), as outlined in Table 2.

2.4.2 Consequence analysis of accidents

In the DES simulation of NATM tunnel construction, representing process durations with fixed values is impractical due to conventional delays. Probability distributions and MC sampling are essential to account for this uncertainty. Distribution fitting, using EasyFit software (version 5.5) in this study, identifies the best-fitting probability distribution based on field-collected data. The identified distribution was then used in AnyLogic's MC module and distribution library to generate random discrete values for simulation. To focus on the central 90% probability range, values within the 5th and 95th percentiles of the cumulative distribution function were retained. This study assumes an absolute error $\omega < 5$, a confidence level $\beta = 95\%$, and 2500 simulation iterations.

In tunnel construction risk management, a comprehensive analysis that addresses both the likelihood and consequences of potential accidents is crucial. Analysis of engineering failure statistical databases and scholarly research reveals that accidents, while critical, are rare within the construction process. The occurrence probability of such accidents varies exponentially across different vulnerability levels. Referring to relevant studies, we utilize the probability scale (Liu et al., 2015; Yu et al., 2017), as shown in Table 3. This scale offers verbal and numerical representations of accident probabilities, making it ideal for assessing low-probability events. Probability ranges are categorized by vulnerability levels, with specific values determined through a linear transformation of assessment scores.

Accidents in tunnel construction significantly disrupt progress, often causing construction stops and project delays. Sousa and Einstein (2012) analyzed 64 tunnel construction accidents resulting in delays of over 60 d. Extending this work, Špačková et al. (2013) included delays of 15–60 d and applied a shifted exponential distribution to model the cumulative probability distribution function F_D for delay times, representing the probability that a delay will not exceed x days, as illustrated in Eq. (20).

$$F_D(x, \lambda, d) = 1 - \exp\left[-\lambda(x - d)\right], \tag{20}$$

where λ is the parameter of the shifted exponential distribution, determined to be 0.0062 via maximum likelihood estimation. The shift parameter d represents the minimum delay duration considered significant. Given that delays of over half a month are viewed as substantial in practical engineering, d is set at 15 d (Špačková et al., 2013).

Consequently, in an MC simulation, each construction cycle (inner loop) is sampled based on the occurrence probability of specific risk events. This sampling assigns labels indicating "whether a risk event occurs" with the Boolean number B_e being 1 (occurrence) or 0 (non-occurrence). When an accident happens, the probability density function f_D, denoting the probability of a delay lasting x days, is derived from the cumulative distribution F_D through differentiation, as shown in Eq. (21). The delay duration is then determined by additional sampling.

$$f_D(x, \lambda, d) = \begin{cases} 0 & B_e = 0 \\ 1 - \exp\left[-\lambda(x - d)\right] & B_e = 1 \end{cases}. \tag{21}$$

3 Case study

3.1 Project overview

The study focuses on a NATM tunnel in Yunnan Province, China, designed as a dual-tube road tunnel, spanning from K9 + 495 to K17 + 150 over 7560.4 m, with elevations ranging from 850 to 1850 m and a relative height difference of 1000 m. Its deepest point is about 933.26 m underground. The most challenging geological conditions are found in the syncline core segment between K10+800 and K15+800, chosen for in-depth analysis due to its complex stratigraphy. The estimated construction time for this section is 1794 d.

Table 2
Evaluation criteria.

E_f	Parameter	Unit	Level			
			I	II	III	IV
Rw	Q_{max}	L/s	>10	1–10	0.1–1.0	<0.1
Sp	Q_s	L/(min·10 m)	>100	25–100	10–25	<10
Wi	K_f	–	<0.4	0.4–0.8	0.8–0.9	0.9–1.0
Fs	K_v	–	<0.15	0.15–0.50	0.50–0.75	>0.75
Df	N_j	–	>3	>3	2–3	1–2
	D_j	m	<0.2	0.2–0.4	0.4–1.0	>1.0
St	T_{st}	d	<1	1–7	7–30	>30
Rp	R_c	MPa	<20	20–30	30–40	>40

Table 3
Accident probability assessment scale.

Grade	I	II	III	IV
Verbal expression	Large	Medium	Small	Improbable
Score	0–0.75	0.75–1.50	1.50–2.25	3–4
Probability scale	0.01–0.10	0.001–0.010	0.0001–0.0010	0.000 01–0.000 10

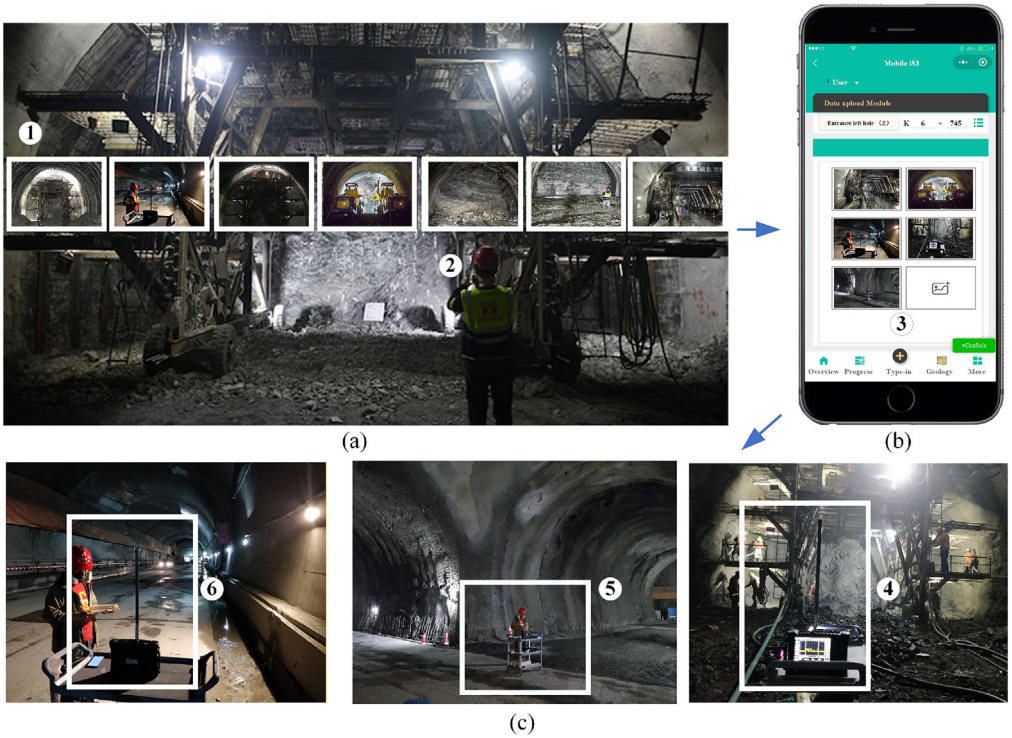
Table 4
Geologic, physical, and mechanical indicators.

Serial	Lithology (Chronology)	ρ (g/cm^3)	φ (°)	f_{rk} (MPa)	f_{ak} (kPa)	q_{sa} (kPa)
①	Boulder (Q_4)	2.30	40	–	400	140
②	Limestone (P_1)	2.69	–	52.8	2500	–
③	Shale (D_2)	2.74	–	16.7	600	140
④	Shale (S_2s)	2.70	–	13.3	600	140
⑤	Argillaceous limestone (S_2s)	2.71	–	26.3	1500	–
⑥	Siltstone (S_1l)	2.65	–	25.0	1200	–
⑦	Shale (S_1l)	2.64	–	22.0	600	140

Preliminary surveys identified seven major engineering geological units, mainly composed of limestone and shale, with the limestone being partly argillaceous. These formations span from the Lower Silurian Longmaxi Formation (S_1l) to the Lower Permian (P_1) and are mostly moderately weathered. Key physical and mechanical properties, such as natural density (ρ), internal friction angle (φ), uniaxial saturated compressive strength (f_{rk}), foundation bearing capacity (f_{ak}), and pile side soil friction resistance (q_{sa}) were collected, with statistical data summarized in Table 4.

As stated in Section 2.1, we deployed the perception and transmission layers of the DT at the construction site to enable effective perception of PE and establish a CN from PE to VE, as depicted in Fig. 11.

The system integrates various sensors installed within the tunnel and handheld devices used by workers to enable



①: On-site image data collection; ②: Field personnel utilizing handheld data acquisition devices; ③: Data uploading via a WeChat Mini Program; ④: Excavation surface transmission node; ⑤: Lateral passageway transmission node; ⑥: In-tunnel transmission node.

Fig. 11. Perception and transmission of physical entity information. (a) Real-time data collection, (b) data uploading, and (c) data transmission nodes.

real-time data collection, as illustrated in Fig. 11(a). The collected data, including environmental parameters, construction progress, and equipment status, provides a comprehensive overview of site dynamics. Multimedia data (e.g., voice, video, and images) and related information are uploaded to a WeChat mini program (Hao et al., 2018), allowing personnel to view, retrieve, and analyze the data anytime, as shown in Fig. 11(b).

Wireless transmission nodes are strategically placed at key locations, such as the tunnel face, connecting passages, and every 1 km along the tunnel, ensuring stable and continuous data transmission, as shown in Fig. 11(c). Leveraging 5G technology, the system rapidly transmits data to wireless nodes and then to the engineering command center, enabling timely and informed decision-making.

3.2 Model setup

3.2.1 Random field setting

The analysis begins by examining geological uncertainties, with the module implemented in Python 3.9.12 using Jupyter Notebook 6.4.8. Similarly, the tunnel vulnerability assessment module was also developed within this environment. For structural simplification and computational efficiency, a rectangular segment with elevations from 830 to 1370 m was chosen for MRF modelling, as shown in Fig. 12. The profile, measuring 5000 m in length and 540 m in width, was discretized into grid units of 15.0 m × 15.0 m, resulting in an MRF grid of 333 rows and 36 columns, or 11 988 units in total. For clarity, the grid is displayed at a resolution of 60.0 m × 60.0 m. Data from 11 borehole logs, collected during the survey phase, are analyzed within this two-dimensional profile.

Leveraging borehole logging data (i.e., assessing the depth of each soil/rock layer across all boreholes) along with a comprehensive understanding of the local geological context obtained by surveyors, we delineated the overall trend of stratum dip angles. This trend has been depicted through a series of refined graphical curves, as shown in Fig. 12. Subsequently, we adopted a distance-weighted interpolation method to calculate the estimated dip angle values φ_{int} for each unit within the entire stochastic field, as illustrated in Eq. (22).

$$\varphi_{int} = \frac{\sum_{i=1}^{8} \omega_i \cdot \varphi_i}{\sum_{i=1}^{k} \omega_i}, \tag{22}$$

where ω_i denotes the weight of the ith neighboring unit, and it satisfies $\omega_i = \frac{1}{d_i}$, where d_i is the distance from the neighboring unit to the central unit; φ_i represents the dip angle value of the ith neighboring unit.

Based on Eq. (22), we calculated the spatial correlation function in each direction. Under the setting $a = 5$, the geological model exhibited a moderately stratified pattern. Considering the central symmetry of the system, four distinct spatial correlation distribution fields were generated, as illustrated in Fig. 13.

3.2.2 Evaluation model construction

To tailor the tunnel vulnerability evaluation model to the case study's specific context, we determined factor weights through expert consultation. Fifteen experts contributed, comprising 10 in tunnel engineering, 3 in urban traffic engineering, and 2 in geotechnical engineering, drawn from a mix of 9 academic institutions, 2 construction companies, 2 design firms, and 2 client organizations.

Then, the AHP–DEMATEL method was utilized to calculate the weights of the indicators through the following steps:

Step 1: Utilizing expert opinions and the AHP method, we conducted a pairwise comparison among E_f and constructed the judgment matrix $\boldsymbol{J}$ (Table 5). By applying Eqs. (10) and (11), the matrix yielded a calculated $\lambda_{\max}$ of 7.182 and a CR of 0.023, indicating an acceptable level of consistency. Subsequently, using Eq. (12), the relative importance weights W^1 were determined.

Step 2: Utilizing the DEMATEL method, we developed the cause-effect relation diagram and constructed the direct influence matrix $\boldsymbol{A}$ (Fig. 14 and Table 6). By applying Eqs. (13)–(18), the values of D_i, B_i, M_i were calculated. M_i was then normalized to derive the influence weight W^2 (Table 7).

Step 3: By applying $W = W^1 W^2$ and normalizing the result, the final weights $\bar{W}$ were obtained (Table 7).

3.3 Results and discussion

3.3.1 Geological uncertainty analysis

Based on the MRF setup in Section 3.2.1, we conducted 100 iterations to produce geological profile configurations. The energy function's changes during these iterations are captured in Fig. 15, showing an exponential decay charac-

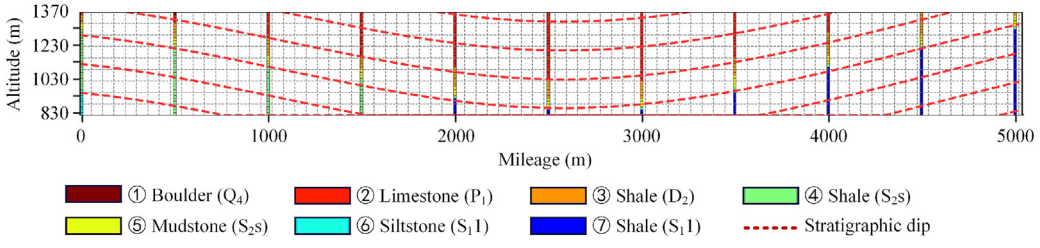


Fig. 12. Profile discretization and borehole data.

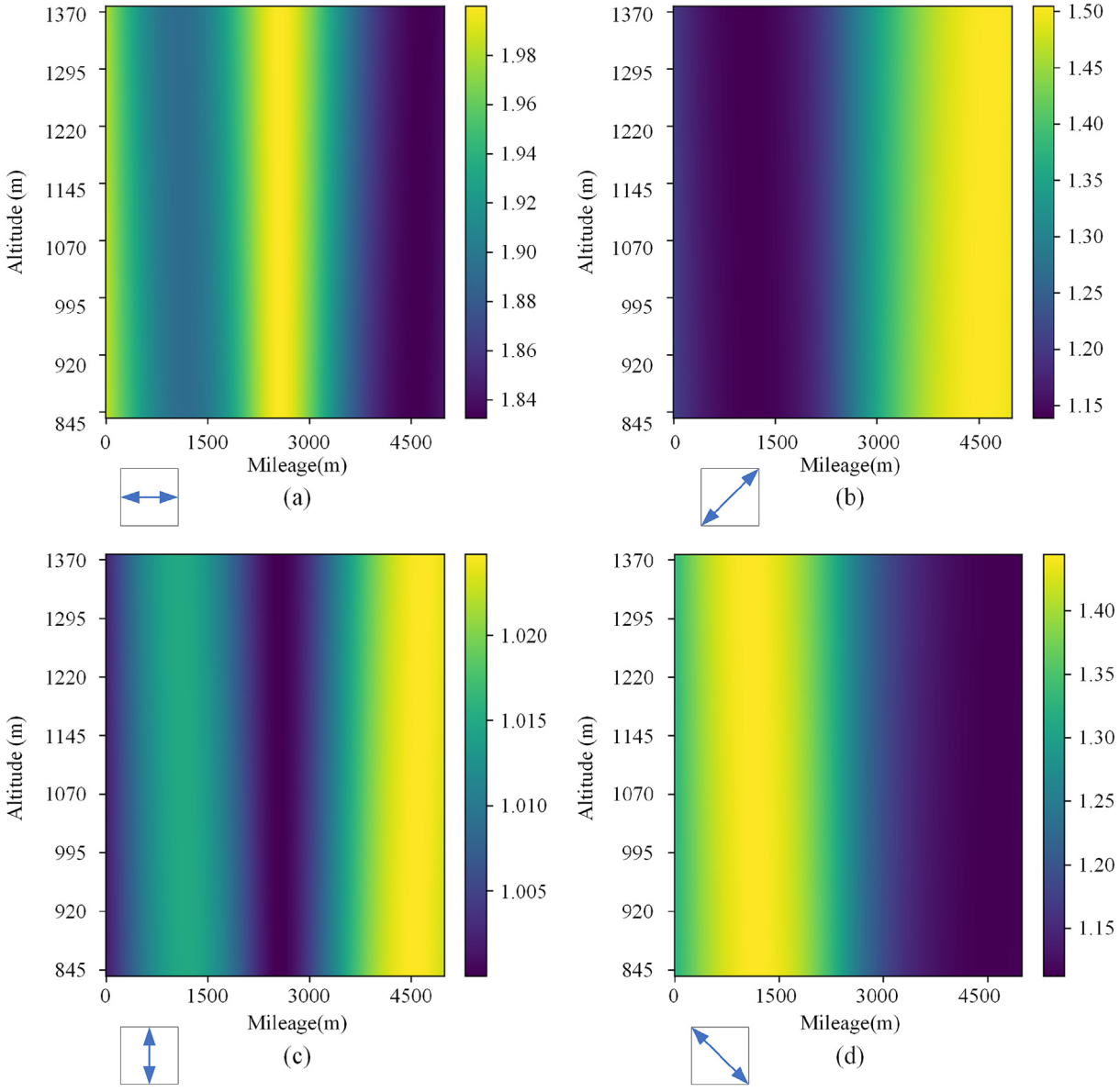


Fig. 13. Spatial correlation distribution field. (a) $\varphi_i = 0°/180°$, (b) $\varphi_i = 45°/225°$, (c) $\varphi_i = 90°/270°$, and (d) $\varphi_i = 135°/315°$.

Table 5
Judgment matrix ***J*** by AHP.

J	Rw	Ss	Wi	Fs	Df	St	Rp
Rw	1	1/3	1/3	1/5	1/7	1/7	1/5
Sp	3	1	1	1/3	1/5	1/5	1/3
Wi	3	1	1	1/3	1/5	1/5	1/3
Fs	5	3	3	1	1/3	1/3	1
Df	7	5	5	3	1	1	3
St	7	5	5	3	1	1	3
Rp	5	3	3	1	1/3	1/3	1

terized by a swift decline in the early stages, which then slowed and stabilized post-60 iterations. The stages of energy reduction (beyond 30 iterations) and stabilization (after

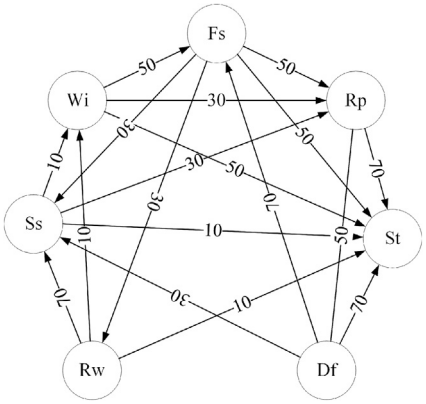


Fig. 14. Cause-effect relation diagram.

100 iterations) are illustrated in Fig. 16(a) and (b), respectively.

After 40 iterations of the MRF model, the geologic profiles generally aligned with borehole data and predefined inclinations. However, local anomalies like sharp corners and discontinuities emerged, indicating suboptimal energy function states. To address these and preserve the geological profiles' overall configuration, the MRF iterative process continued until stability was achieved. The refined model, shown in Fig. 16(b), effectively corrected most local discrepancies, maintaining the intended stratum inclinations and achieving a synclinal geomorphology, with strata flanking upwards and a central downward curve. Notably, at the 2000 m mark, a pronounced strata transition marked by a fault was accurately captured.

A parameter sensitivity analysis was conducted to examine the influence of the spatial correlation intensity parameter a on geologic profile modelling, as illustrated in Fig. 17. When spatial correlation is not considered ($a = 0$), the simulated stratigraphic attributes exhibit a completely random distribution, resulting in a disordered and geologically meaningless configuration. As a increases from 1 to 3, the profile demonstrates a clear and systematic transformation. The stratigraphic patterns gradually evolve into well-defined sedimentary sequences, with a significant reduction in isolated anomalous color patches, and stratigraphic boundaries becoming smoother and more continuous. By $a = 5$, as shown in Fig. 16(b), the geologic profile reaches an optimal state, with clearly layered and continuous strata in the synclinal core. This progressive improvement clearly demonstrates the effectiveness of incorporating a spatial correlation field as a constraint within the MRF method.

Following the creation of numerous geologic profiles via the modelling process, the concept of information entropy H_i was utilized to quantify the uncertainty of these outcomes, a method introduced in the realm of geological modelling by Wellmann and Regenauer-Lieb (2012). For each unit within the MRF, H_i is determined according to Eq. (23):

$$H_i = -\sum_{j}^{n} p_j \log p_j, \tag{23}$$

Table 6
Direct influence matrix $\boldsymbol{A}$ by DEMATEL.

$\boldsymbol{A}$	Rw	Ss	Wi	Fs	Df	St	Rp
Rw	0	70	10	0	0	10	0
Sp	0	0	10	0	0	10	30
Wi	0	0	0	50	0	50	30
Fs	30	30	0	0	0	50	50
Df	0	30	0	70	0	70	50
St	0	0	0	0	0	0	0
Rp	0	0	0	0	0	70	0

Table 7
Comprehensive weights of evaluation factors.

E_f	AHP-results		DEMATEL-results				$\bar{W}$(%)
	ω^1	W^1 (%)	D_i	B_i	M_i	W^2 (%)	
Rw	0.27	2.76	0.55	0.21	0.76	8.77	1.47
Sp	0.54	5.63	0.31	0.74	1.05	12.07	4.13
Wi	0.54	5.63	0.84	0.13	0.98	11.26	3.85
Fs	1.26	13.13	0.92	0.58	1.49	17.20	13.72
Df	2.86	29.86	1.41	0.00	1.41	16.21	29.41
St	2.86	29.86	0.00	1.70	1.70	19.57	35.51
Rp	1.26	13.13	0.32	0.98	1.30	14.93	11.91

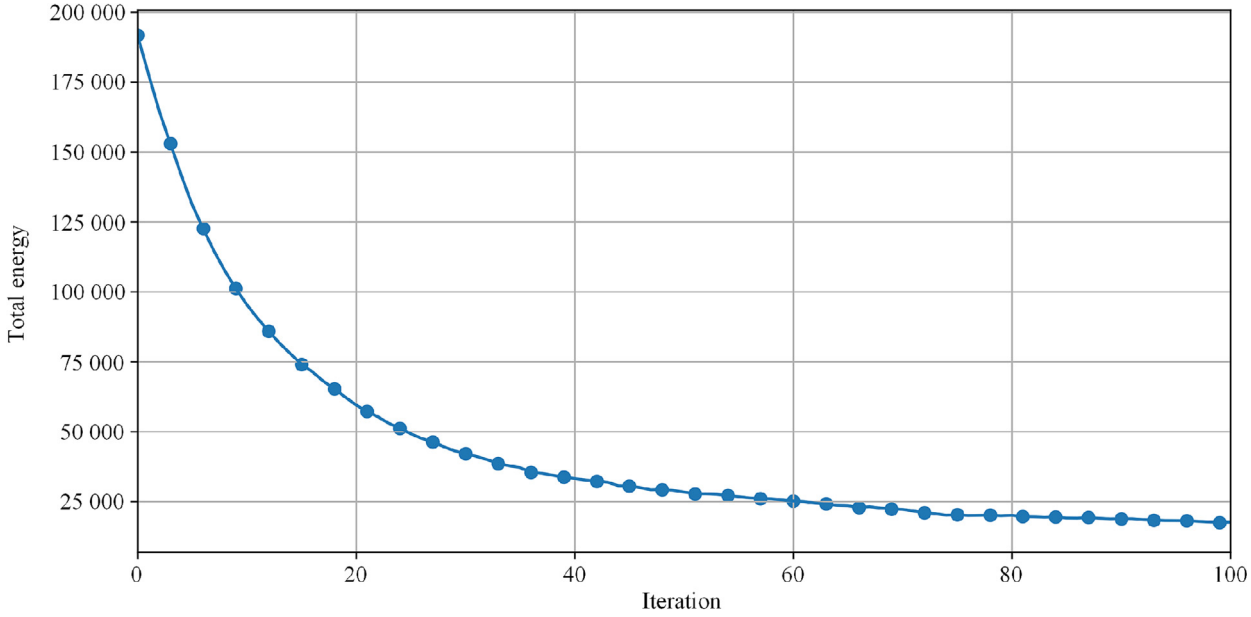


Fig. 15. Energy variation curve with iterations.

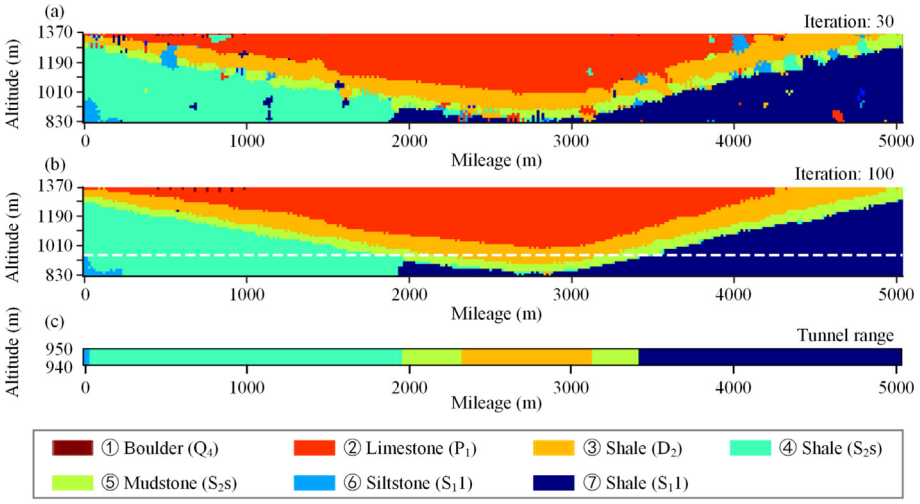


Fig. 16. MRF simulated results. (a) Geologic profile with 40 iterations, (b) & (c) geologic profile and tunnel profile with 100 iterations.

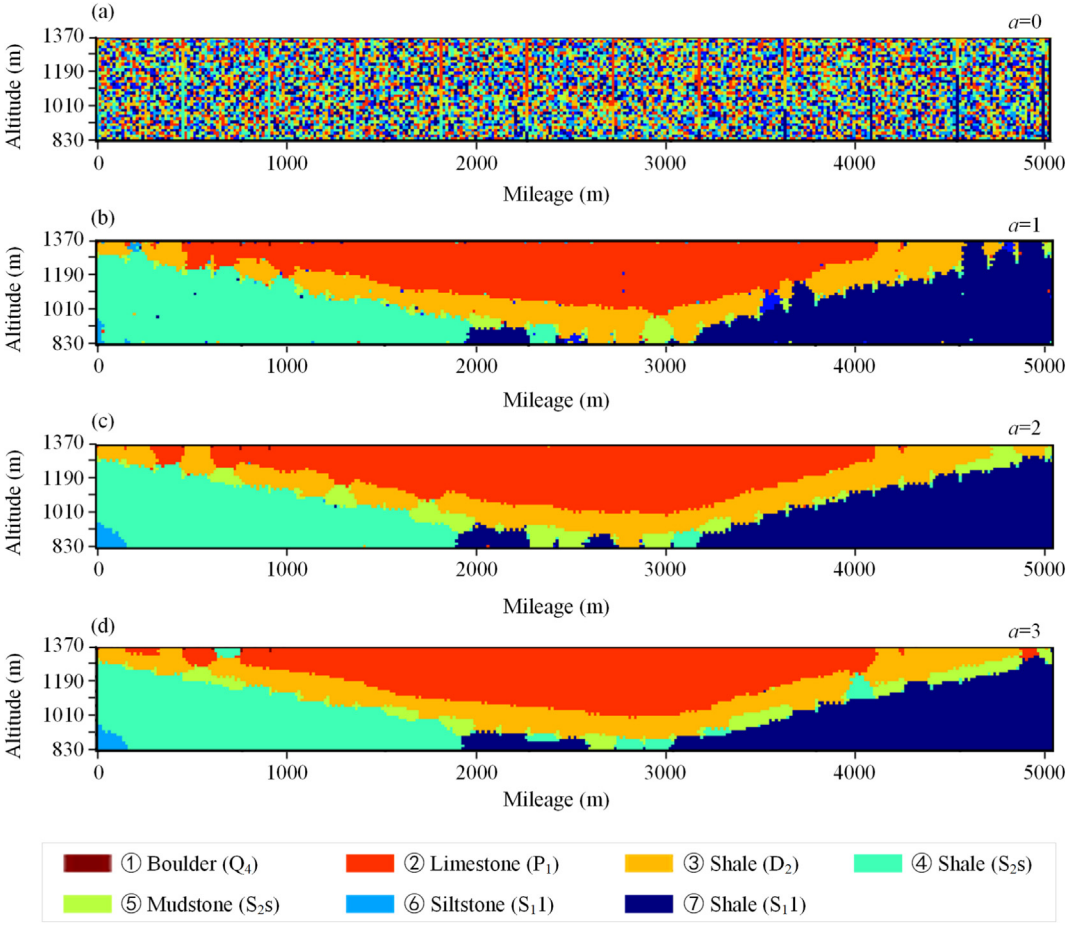


Fig. 17. Sensitivity analysis results of the spatial correlation intensity parameter.

where n is the count of potential geologic labels for a unit, with p_j representing the probability of each label. A unit's information entropy H_i value, which is higher when the label's uncertainty increases, serves as a measure of this uncertainty. The normalized information entropy map in Fig. 18 shows that uncertainty in units near borehole locations is minimal, thanks to the direct influence of borehole data. However, units farther from boreholes experience greater uncertainty, especially at the edges of geologic boundaries. This indicates a significant uncertainty for

units at geologic interfaces, as their labels often change throughout the iterative process before reaching energetic stability, thereby delineating clearer geologic boundaries.

At the tunnel's axial elevation (~925 m), geologic data from the regional geological profile provided the basis for assessing tunnel accident vulnerability, depicted in Fig. 16 (c). Statistical analysis of geologic changes along the elevation gradient, shown in Fig. 19, reveals that tunnel construction will encounter five main strata types. Predominantly, Shale (S_2s), coded as ④, and Shale (S_1l), coded as ⑦, comprise 70.2% of the strata, offering relatively intact conditions and favorable construction properties. However, Shale (D_2), coded as ③, and Argillaceous limestone (S_2s), coded as ⑤, make up 13.2% and 15.9% of the stratigraphy, respectively, located in the core of the syncline (2000–3000 m elevation). These layers experience compressional stress, leading to fragmentation and diminished self-stabilization, thus presenting higher construction risks.

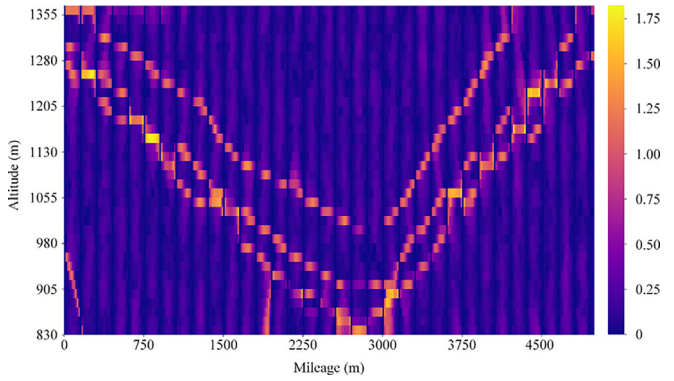


Fig. 18. Geologic uncertainty visualization via normalized information entropy.

3.3.2 Construction-related uncertainty

Utilizing the geologic distribution from MRF modelling, along with data on physical and mechanical properties, hydrogeology, engineering geology, and surrounding rock stability from surveys, geophysical explorations, and lab tests, we applied the scoring criteria from Table 2 to evaluate factors along the tunnel axis, as shown in Fig. 20. The tunnel's deep burial means it traverses a moderately weathered zone throughout, leading to a uniform score of 1 for the factor W_i.

After conducting weighted calculations based on Table 7, the vulnerability levels of tunnel accidents were ascertained, as depicted in Fig. 21. The distribution across levels I–IV is 25.15%, 16.47%, 22.46%, and 35.92%, respectively, highlighting that the lowest risk sections (level IV) constitute the largest portion, followed notably by the highest risk sections (level I). Specifically, the segment between 2600 and 3000 m was identified as the highest risk, aligning with field survey findings. This section lies at a syncline core, marked by significant groundwater development and fragmented rock, underscoring the evaluation model's accuracy.

To assess conventional delay factors, the team tracked construction performance meticulously over 15 d, utilizing stopwatches and measuring tapes for precise measurement. Data pertinent to the engineering processes were recorded

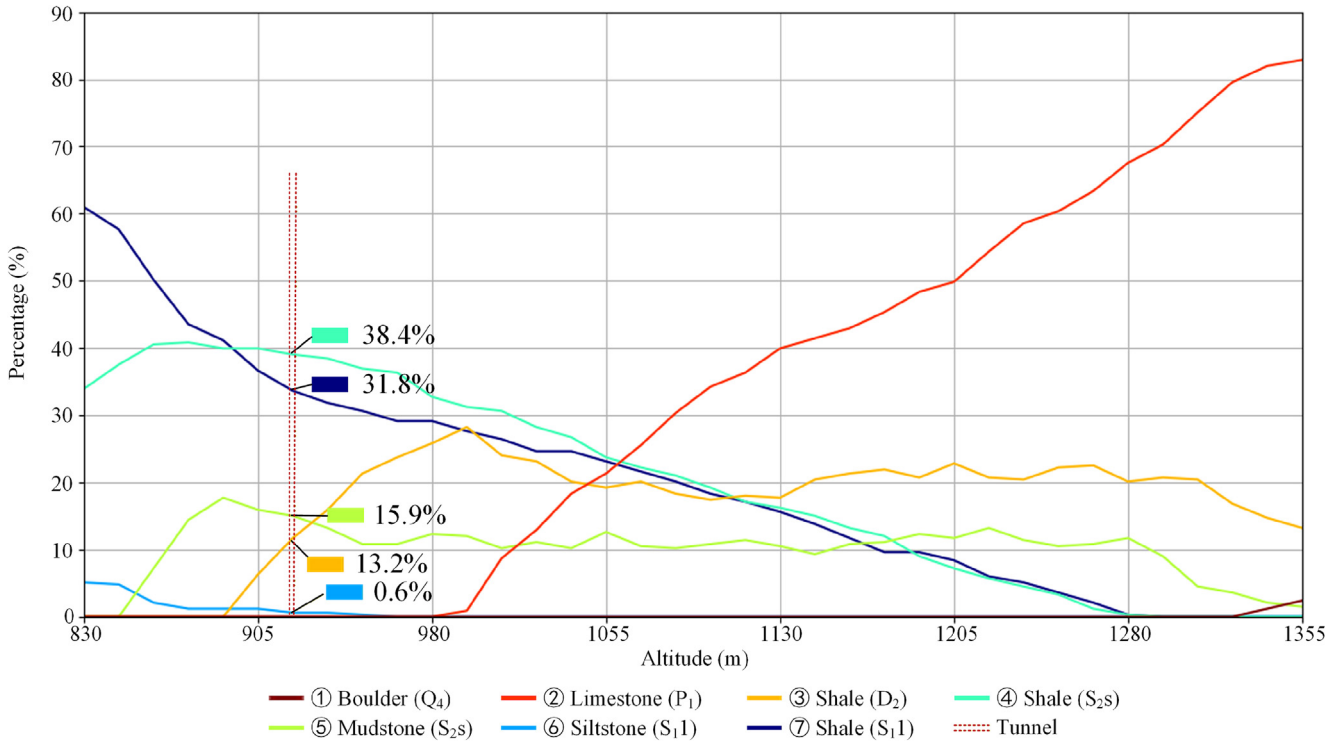


Fig. 19. Changes in strata distribution along the elevation gradient.

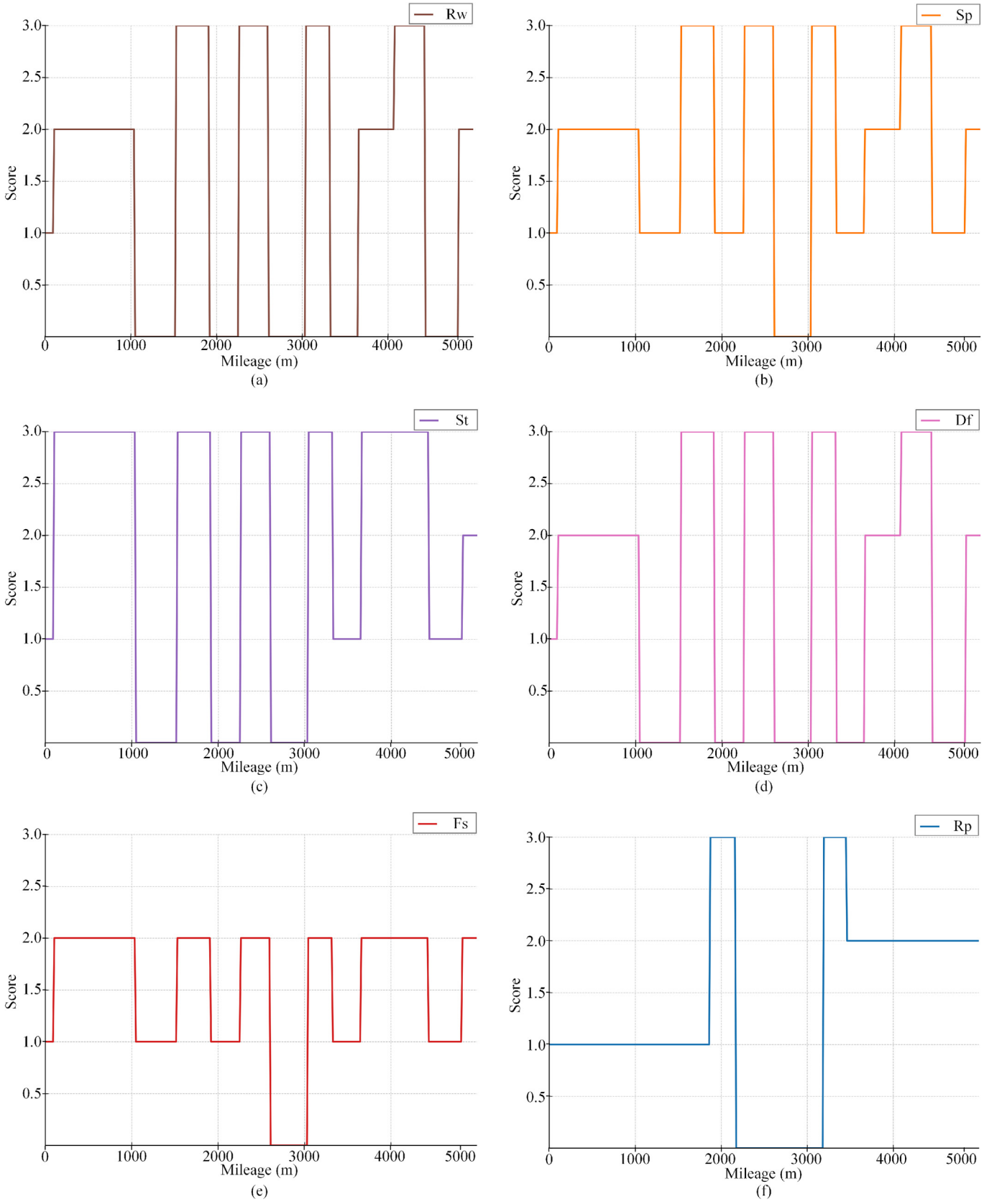


Fig. 20. Variation diagram of indicator scores. (a) Rw, (b) Sp, (c) St, (d) Df, (e) Fs, and (f) Rp.

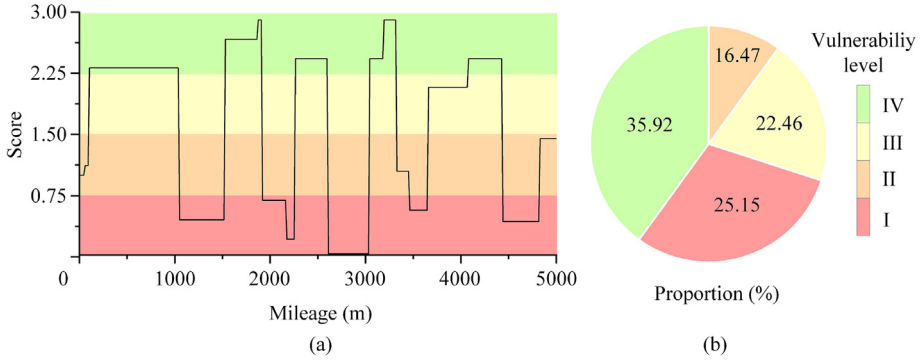


Fig. 21. Risk assessment outcomes. (a) Mileage distribution, and (b) proportion of each level.

and uploaded via a WeChat mini program, detailing tunnelling lengths and times for critical construction activities. These data, essential for the simulation model's inputs, necessitated the selection of appropriate probability distributions to simulate process times accurately. AnyLogic software, which offers a default set of probability distributions for simulation, was used alongside EasyFit software to analyze the data and identify the most fitting probability distribution curve (Schittkowski, 2002). The resulting probability distribution functions' parameters, including mean (μ), median (M), standard deviation (σ), skewness (S_k), and kurtosis (K_u), are compiled in Table 8.

Table 8 indicates that construction sections with low vulnerability levels III and IV are in favorable geological conditions, with the surrounding rock exhibiting strong self-stabilizing properties. Thus, advanced support and supporting structures (e.g., steel frames, anchor rods, steel mesh) are often unnecessary in these areas. Except for the constant duration of setting out (including ventilation) operations, other construction activities tend to show a positive skewness distribution, indicating higher construction efficiency with occasional delays due to conventional factors. The K_u of these distributions is greater than 0 but remains low (less than 2), suggesting a concentration of operation durations around the mean with few extreme outliers. However, it is important to note that, due to the uncertainty in the stratum-structure response, segments requiring structural construction (such as the advanced support, concrete, and support structure) show higher K_u and S_k values, highlighting greater variability and uncertainty in these phases and underscoring the need for meticulous attention to structural construction activities.

Table 8
Probability distribution functions to model productivity and logistics processes.

Steps	Vulnerability level	Probability distribution	Parameters				
			μ	M	σ	S_k	K_u
Cyclic advance	I	Normal	79.82	80.00	2.74	0	0
	II	Normal	201.24	200.00	4.31	0	0
	III & IV	Normal	249.73	250.00	5.52	0	0
Survey	I–IV	Gamma	32.41	36.74	17.32	0.72	0.32
Setting out	I–IV	Constant	30.00	30.00	0		
Advance support	I & II	Weibull	66.21	62.60	18.07	1.10	1.49
Drilling & Explosion	I & II	Lognormal	101.86	101.06	7.26	0.70	0.87
	III & IV	Lognormal	185.13	183.78	9.99	0.88	1.40
Safety inspection	I–IV	Gamma	55.02	51.74	17.32	1.15	0.43
Debris removal	I	Weibull	58.29	57.49	6.95	0.63	0.25
	II	Weibull	87.72	86.65	9.27	0.63	0.25
	III & IV	Weibull	178.06	178.47	4.72	0.46	0.19
Primary concrete	III	Gamma	58.16	62.33	4.23	0.92	1.40
	I & II, IV	Gamma	31.32	32.45	2.31	1.15	1.32
Supporting structures	I	Lognormal	123.29	122.20	4.22	0.77	1.32
	II	Lognormal	183.29	182.20	7.40	0.98	1.77
Secondary concrete	I & II	Gamma	58.04	61.74	17.32	1.15	1.60
	III & IV	Gamma	32.91	29.88	8.13	0.78	1.83

In practical engineering applications, the spatial distribution of construction segments across different vulnerability levels is often highly heterogeneous. To examine the relationship between PSM simulated results and vulnerability levels, a sensitivity analysis was conducted. In this analysis, the influence of accidents was excluded, and 1000 MC simulations were performed for each vulnerability level. Figure 22 illustrates the simulated construction cycle durations, while their statistical distributions are presented as histograms in Fig. 23.

The mean cycle durations for levels I–IV were 555.9, 610.6, 581.3, and 519.8 min, respectively, with corresponding standard deviations of 21.8, 25.2, 22.4, and 23.6. Notably, level II segments exhibited the longest average cycle duration and the greatest variability. This outcome is primarily attributed to their extended excavation lengths (averaging 201 cm) and more stringent safety control requirements. Conversely, despite being designated as the highest vulnerability category, level I segments demonstrated a shorter average cycle time. This counterintuitive result is explained by their notably shorter excavation lengths, with a mean of only 79.8 cm, representing just 39.7% of level II and 31.9% of levels III and IV. These findings indicate that the simulation outcomes effectively capture the construction characteristics under varying vulnerability conditions. The observed variability in construction durations reflects inherent uncertainties associated with conventional construction delays.

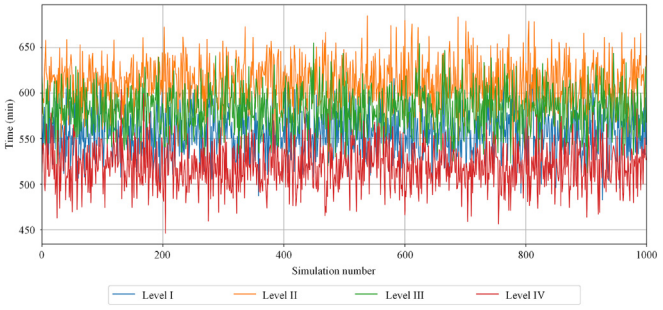


Fig. 22. Simulated durations under different vulnerability levels.

3.3.3 PSM simulation results

Leveraging the accident vulnerability assessment and integrating conventional delay distributions from Table 8, alongside the probability of accident occurrence and potential delays as Eq. (20), a process simulation entailing 2500 MC iterations was performed for the case study tunnel construction. This simulation ran on an AMD Ryzen™ 7 7735H @ 4.75 GHz processor featuring eight cores and sixteen threads, with 16.0 GB of RAM, completing in 621.972 s.

Figure 24 shows that against the baseline construction schedule of 1794 d estimated by the design unit, 2500 MC simulations predicted the shortest and most ideal tunnel excavation time as 1453 d, marking a 19.01% time saving. The average projected duration was 1740 d, closely matching the baseline with a mere 3.01% variance. This reflects the model's accuracy compared to expert estimates and implies conventional delays might add about 287 d (16.00%) to the ideal scenario. The longest projected duration was 2192 d, suggesting that delays and accidents could extend the project by up to 398 d (22.19%) over the baseline, or 739 d (41.2%) beyond the ideal case.

Figure 25 presents a histogram of construction durations with an overlaid cumulative distribution curve, showing the completion probability within certain timeframes.

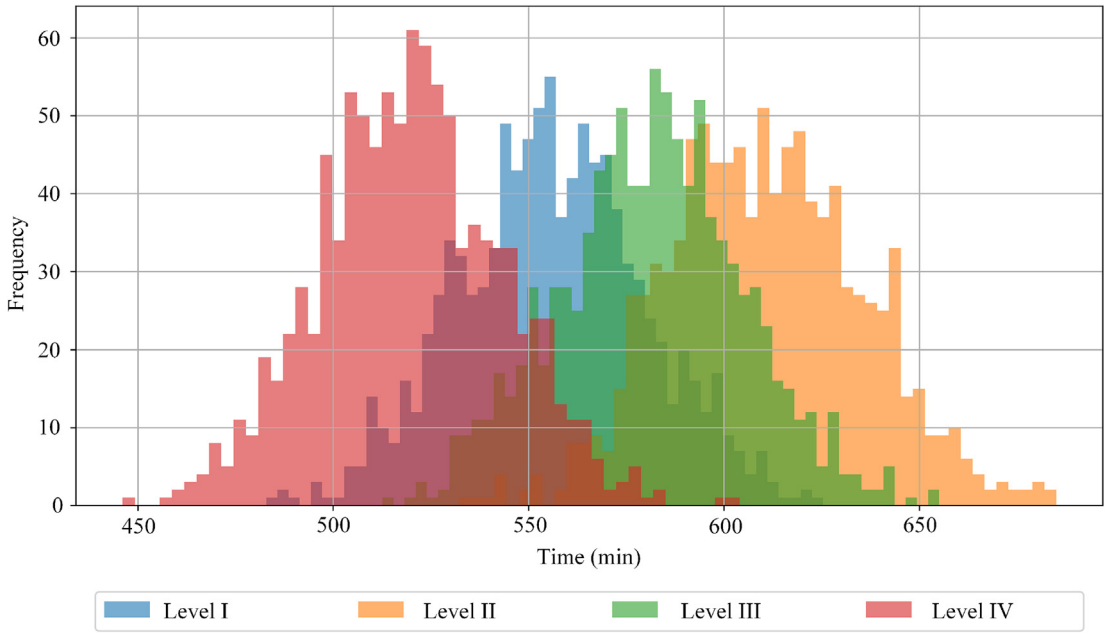


Fig. 23. Histogram of the durations under different vulnerability levels.

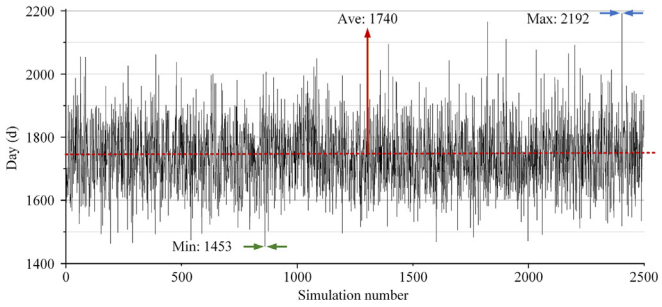


Fig. 24. Construction duration obtained from the process simulation.

The simulation suggests a 78.4% chance of finishing within the estimated 1794 d. Statistical analysis reveals an 18.98% likelihood for completion between 1450 and 1650 d, a 67.00% chance within 1650–1850 d, and a 14.02% probability for the longest span of 1850–2200 d, as detailed in Fig. 26(a). The histogram's right-side long tail and the cumulative curve's initial steep rise, flattening out later, indicate most projects cluster around the mean duration but with potential for significant delays. This pattern underscores the impact of low-probability accidents on extending construction timelines, highlighting their substantial influence on schedule risk.

In an analysis of 146 simulations out of 2500 that encountered accidents, depicted in Fig. 27, delays ranged from 15 to 512 d due to accidents. Statistical insights from Fig. 26(b) and (c) show these incidents resulted in a total of 227 326 construction days, with 25 337 d directly attributable to accidents, constituting 11.1% of the overall duration. This underlines the substantial impact of accidents on construction timelines. Moreover, accident frequencies in segments of vulnerability levels I–III were 132, 11, and 3, respectively, equating to 90.41%, 7.53%, and 2.06% of accidents, with level IV segments experiencing no accidents. These findings emphasize the need for enhanced monitoring and early warning in segments with a higher risk to mitigate accident occurrences.

3.4 System implementation and verification

The case project personnel are divided into teams specializing in design, construction, management, supervision, etc. Each team is granted access to the DT-enabled, iS3-based system, along with interface links and credentials. As outlined in Section 2.1, the system functions as the alternation layer within the DT framework, delivering the Ss, including enabling extensive information interaction via web interfaces and WeChat mini programs. Additionally, it acts as the CN, linking the VE, which comprises digital and analytical models, to the PE, which includes on-site elements, workers, and remote managers. The system provides real-time access to construction vulnerability levels, progress, and deviations from the DES model.

The web interface, as illustrated in Fig. 28, supports visualization of the as-built tunnel structure and geological models. Key construction stages such as the excavation face, secondary lining, and inverted arch are highlighted with visual indicators. The analysis module monitors progress by section and compares it with the simulation model to identify deviations. The vulnerability module evaluates construction risks within 40 m ahead of the excavation face. Results are updated every 30 min based on real-time progress. The WeChat mini program shown in Fig. 28 provides mobile access to progress information and vulnerability analysis results for field workers. It also offers a query

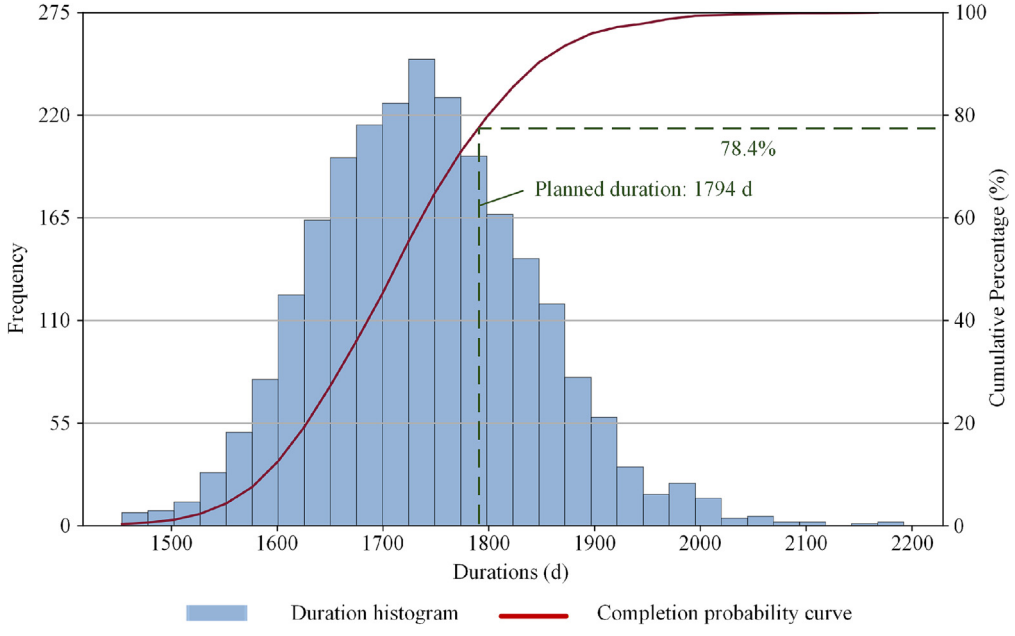


Fig. 25. Cumulative probability curve for the tunnel construction duration.

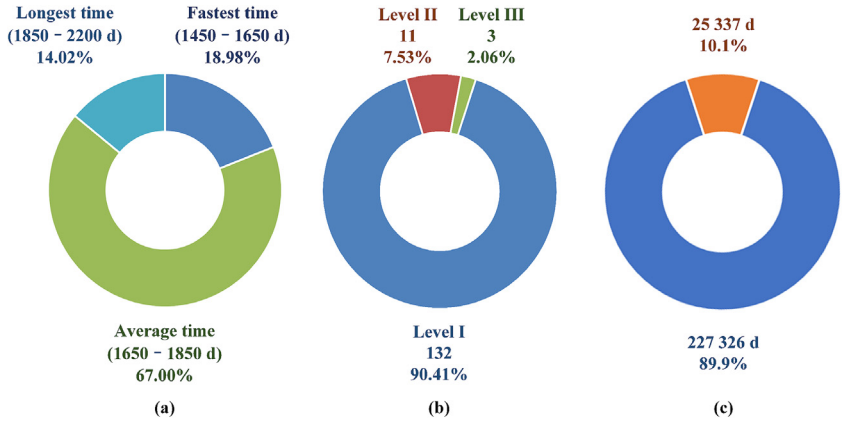


Fig. 26. Statistical analysis of the impact of accidents. (a) Duration statistics, (b) relationship between accidents and vulnerability levels, and (c) accident-induced delay time.

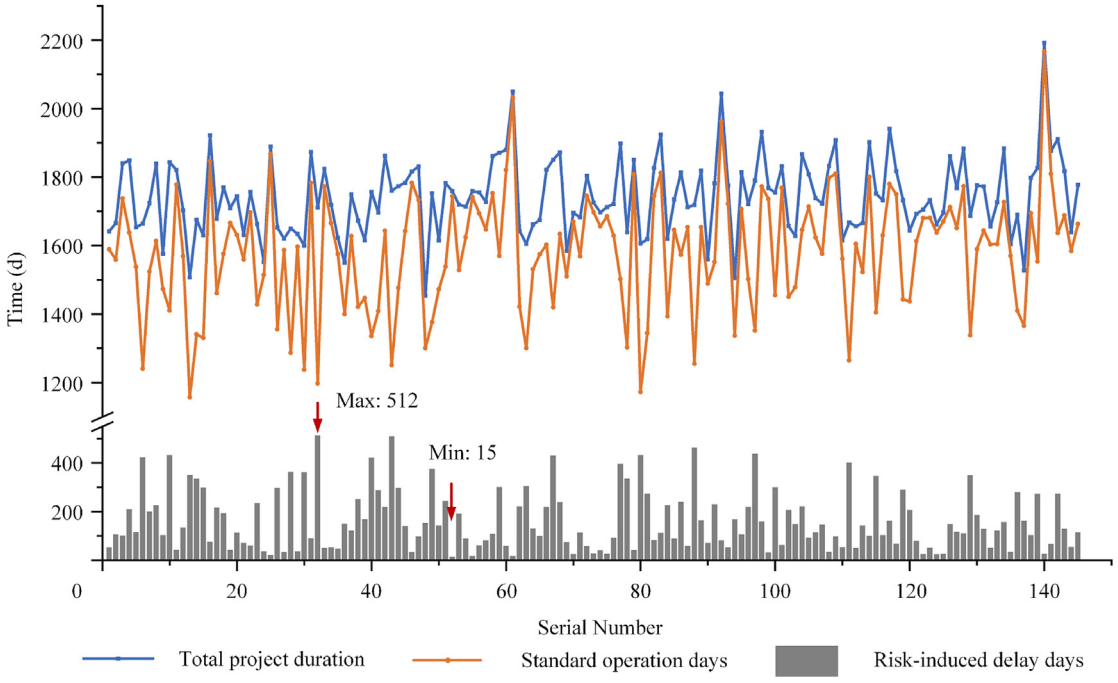


Fig. 27. Simulation results affected by accidents.

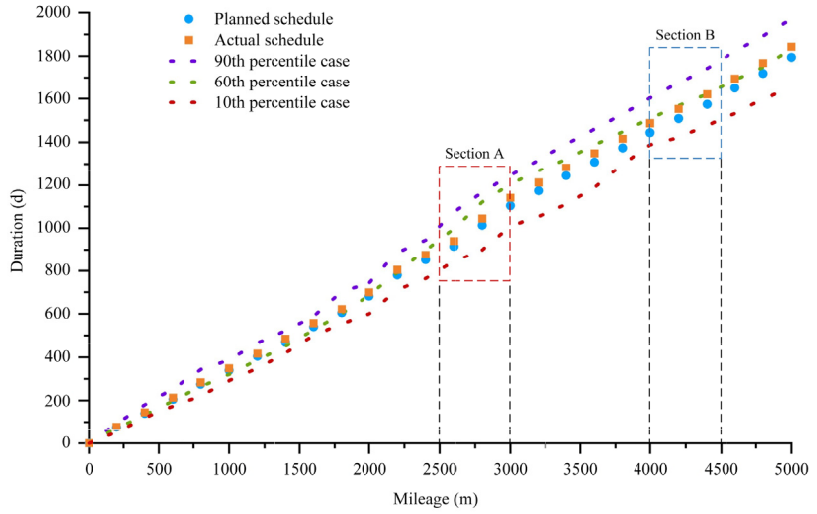


Fig. 28. Main interfaces for web and mobile interaction.

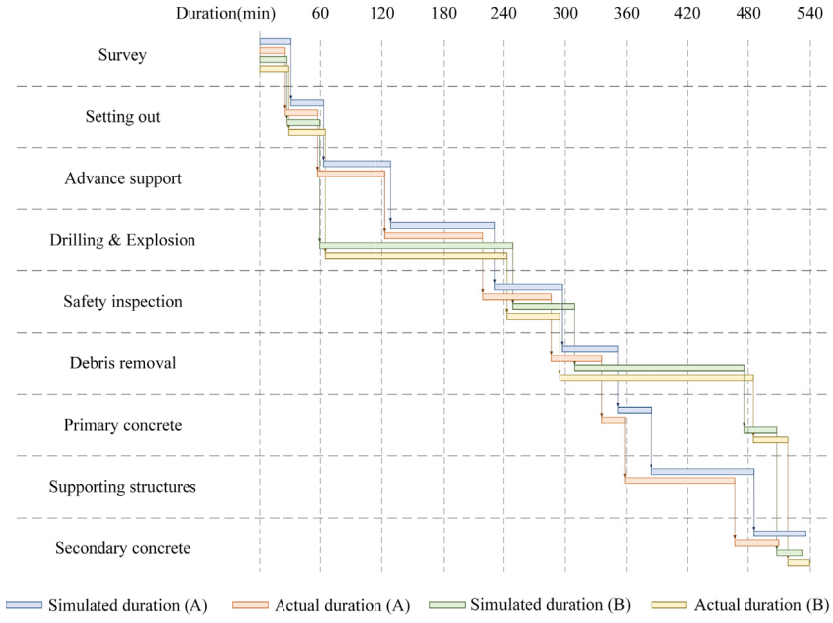
function for geological categories and related parameters at the current construction site.

Compared to traditional multi-party communication methods, the DT-enabled system enhances efficiency, accuracy, and reliability. Traditional approaches rely on paper reports (3–4 per day with 180 min delays) and face-to-face meetings (twice daily with 360 min delays). In contrast, the web and mobile interfaces refresh every 30 s, support 2 MB/s upload and 10 MB/s download speeds, and enable real-time data exchange across multiple project units. Furthermore, built on a microservice architecture, the system supports flexible integration of sub-services developed in. Net, Python, Java, and other languages. A scalable data service framework facilitates access to heterogeneous data sources, along with distributed storage and concurrent processing. In high-concurrency environments, it supports the processing of over 10 000 data streams simultaneously, with minimal latency (<100 ms). Additionally, a multi-threaded simulation engine and incremental update mechanism enable efficient iterative simulations. Architectural and implementation details are provided in the authors' prior research (Li et al., 2024b). The system ensures accuracy and reliability through automated verification and real-time updates, significantly minimizing delays in data collection, dissemination, and decision-making.

The DT-enabled system comprehensively recorded the actual construction progress of the case section (K10 +800 to K15+800), serving as a basis for validating the



(a)



(b)

Fig. 29. Comparison of schedules from the simulation model and real site. (a) Construction durations, and (b) Gantt chart of excavation cycles.

PSM model. As illustrated in Fig. 29(a), the PSM model outputs were compared against both the actual and the planned schedules provided by the design unit. Based on 2500 simulation runs, construction durations were ranked from shortest to longest, and percentile curves at the 10th, 60th, and 90th percentiles were plotted, representing the optimal, average, and pessimistic scenarios, respectively.

The comparison results indicate that the actual construction duration generally exceeded the planned schedule. However, the average percentile curve closely matched the actual progress in most segments, suggesting that the PSM model incorporated common delays more comprehensively than traditional scheduling methods. In the syncline core section between 2500 and 3000 m (Section A in Fig. 29(a)), the geological model identified adverse conditions with high accident potential. A single excavation cycle from Section A was selected, and its Gantt chart is shown in Fig. 29(b). The PSM model adopted a conservative estimate in this section, assigning longer durations to individual tasks. This led to a cumulative delay of 28 min compared to the actual schedule. The simulated average excavation rate in this section was 2.48 m/d, whereas the actual rate reached 2.69 m/d. Consequently, the average curve exceeded both the actual and planned curves. The conservative estimation provided a safety buffer for on-site operations and was communicated in real time via the DT platform. After passing through the high-risk zone, construction efficiency improved as site personnel responded to updated risk mitigation measures. This is reflected in Section B of Fig. 29(a), where the Gantt chart of a typical cycle in Fig. 29(b) shows a high degree of alignment with actual durations, completing 13 min ahead of schedule. Ultimately, the PSM model's predicted completion time deviated by only 9 d from the actual duration, representing just 0.50% of the total project timeline. This outcome demonstrates the effective integration of process, geological, and vulnerability data with real-time construction decision-making.

Notably, the actual schedule curve consistently remained within the 10th–90th percentile curves, showing a stable trend with minimal deviations. This confirms that the uncertainty parameters in the PSM model are well-calibrated, preventing significant or contrary deviations even under extreme conditions. The DT-enabled system continuously provided real-time progress tracking and vulnerability alerts throughout the construction process. No accidents occurred during the implementation of this tunnel segment. Compared to the estimated least favorable scenario, the project saved 320 d, equivalent to 17.8% of the total construction period. According to project budget estimates and traffic flow forecasts, each day saved translates into an average cost reduction of approximately ¥560 000 (US$78 200). The route supports a daily traffic volume of 30 115 standard passenger cars. Therefore, accurate process simulation and DT-based project management deliver substantial economic and social benefits to both the project and the local community.

4 Conclusions

A DT-enabled process simulation system is proposed to mitigate the risks of losses and delays arising from multiple-source uncertainties in NATM tunnel construction. The system takes into account geological uncertainties and construction-related uncertainties, including conventional delays and accident recovery durations. By putting the conceptual DT framework into practice, it bridges the gap between physical entities and virtual models and offers recommendations for site management and early warning. The following conclusions can be summarized:

(1) Combining the characteristics of NATM tunnel construction, a system framework was constructed based on the DT concept. This framework is structured into four primary levels. At the perception level, a variety of information collection sources is deployed at the tunnel construction site to comprehensively perceive the physical entities and relevant information. In the transmission level, data interchange from the excavation face to the control center is facilitated through wireless network technology and the deployment of digital transmission nodes. At the modelling level, the creation of a data-driven PSM model is realized alongside visualization modelling and information integration within a virtual space. Finally, at the interaction level, the transmission and coordination management of construction information and analysis results are achieved through multi-end connectivity.

(2) In the modelling layer, the integrated PSM model was proposed. Initially, the SysML was employed for structured modelling of the construction system components and execution logic. Building upon this, a PSM model based on DES was developed using AnyLogic. This model features a dual-loop nested simulation execution method and characterizes the uncertainty of conventional delays using probability density functions specific to each procedure. Subsequently, to comprehensively address uncertainties, the stochastic geological modelling module based on MRF and the module for evaluating the vulnerability of tunnel construction based on AHP–DEMATEL were constructed.

(3) A NATM tunnel project was selected to validate and illustrate the DT-enabled system. The process began with an analysis of geological and construction-related uncertainties. Based on this, 2500 Monte Carlo simulations were conducted, followed by a statistical analysis of the results. The simulation outcomes indicate a 78.4% probability of completing

the tunnel project within the planned duration of 1794 d as estimated by the design unit. The cumulative distribution curve exhibits a positive skewness, suggesting that the expected completion times are concentrated around the mean due to regular delays caused by geological conditions, equipment issues, and worker performance, yet significant delays are also possible. Although the frequency of construction accidents is low, their impact is substantial, potentially extending the overall project duration by up to 11.1%.

(4) The DT system was applied in the actual construction of the case. Initially, an explanation and analysis of the system's multi-end interaction and collaborative management were provided. Subsequently, the simulation results were compared with the actual schedule. The comparison indicates that the average estimated schedule provided by the model closely aligns with the actual situation, differing by only 9 d, which represents a deviation of 0.50%. Furthermore, with the assistance of the system, the actual duration of the project significantly outperformed the most adverse estimated scenario, resulting in enhanced economic and social value.

However, the DT-enabled process simulation system still has some limitations and areas for future work:

(1) Due to the sparsity of geological data, the geological analysis module currently employs a relatively coarse grid resolution, which is insufficient for detailed analyses, such as finite element method simulations. This limitation can be addressed by incorporating construction-phase survey data. Potential improvements include integrating advanced forecasting data (e.g., tunnel seismic prediction and ground penetrating radar), high-precision point cloud data, and utilizing computer vision for automated extraction of rock features from excavation face images (Lei et al., 2024).

(2) On-site construction step durations are still recorded manually. As a result, the limited data available leads to static probability distributions for the PSM model parameters, with fixed distribution types that fail to capture the randomness and complexity introduced by risks. To address this, the DT system could consider deploying 24 h cameras for process detection and automated duration recording, while integrating advanced real-time analysis methods to enable dynamic updates of the PSM model parameters (Eshaghi et al., 2024).

(3) This study explores the integration of DT technology in the NATM field, yet it faces several challenges. The construction of virtual entities in DT should rely on large-scale, high-quality databases. The use of intelligent algorithms to identify, clean, and correct dirty and irrelevant data to improve data quality is a common theme in all big data approaches (Xiao et al., 2022). What's more, existing sensing and monitoring methods lack the necessary level of automation and intelligence, relying primarily on manual subjective measurements, which cannot ensure reliability and timeliness. Future advancements could involve utilizing sensor networks or Internet of Things (IoT) devices to obtain data with high accuracy and long-term stability requirements (e.g., strain and settlement data) (Li et al., 2024a). Furthermore, the DT framework established in this study still represents indirect control over the construction site. Establishing a real-time feedback mechanism is crucial, not only for guiding workers but also for integration into the automated control systems of NATM machinery. The ultimate goal is to achieve unmanned construction, a key direction for future development.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Sicheng Zhao: Writing – original draft, Software, Methodology. **Yi Shen:** Writing – review & editing. **Mei Yin:** Supervision, Conceptualization. **Tao Li:** Investigation, Conceptualization. **Wei Wu:** Investigation. **Xiaojun Li:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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